

When Can a Machine Judge Like a Human? Evaluating LLM-as-a-Judge Approaches for Human-Centered Computing Research

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Large language models (LLMs) are flooding into Human-Centered Computing (HCC) research as substitutes for human judgment: coding qualitative data, annotating crowdsourced corpora, simulating study participants, generating personas, and evaluating designs. Yet HCC’s methods carry epistemological commitments (interpretation, reflexivity, positionality) that make uncritical substitution risky, and the field lacks an evidence-based account of *when* an “LLM-as-a-judge” can stand in for a human and when it cannot. We address this with four studies. **Study 1a** is a PRISMA systematic review of 217 studies across 29 domains that establishes the general reliability landscape of LLM-as-a-judge: substitution is validated ($\approx$ or above the human-human ceiling) for objective, verifiable, low-expertise tasks judged in aggregate, and unreliable for subjective, expert, low-resource, or adversarial ones (mean reliability 3.15/5; only 31% of studies in the reliable band). **Study 1b** asks whether this landscape is a moving target: we re-run a flagship “promising” benchmark (SummEval) with the most powerful open models served via an institutional vLLM API and ask whether they now *surpass* the agreement numbers reported for proprietary judges, distinguishing a “matter of time” from a structural ceiling. **Study 2** is a PRISMA systematic review of 55 studies focused on HCC, HCI, and qualitative-research methods that historically require human input. It finds that the field overwhelmingly positions LLMs to *augment* rather than replace humans (36 augment vs. 14 replace vs. 5 simulate), with an epistemic stance dominated by *tension* (38/55); objective crowd-annotation substitution is validated, interpretive qualitative analysis is reliable only as human-in-the-loop augmentation, and *simulating* participants, personas, and usability evaluations is the least reliable (flattening of identity groups, caricature, hallucinated usability issues). **Study 3** closes the gaps the first two studies expose, by running a standardized LLM-as-a-judge meta-evaluation, using open models served via an institutional vLLM API and the agreement-against-human-ceiling methodology distilled from Studies 1-2, on HCC datasets that contain human labels but have never been formally tested with an LLM judge. Across 18000 judgments (a 6-model open panel, 9B-754B), it finds the human-agreement ceiling governs reliability: the panel matches or exceeds humans on low-ceiling NLI but falls short of, and fails the alt-test for, subjective social-computing annotation (emotion, hate speech), with binary offensiveness the strongest case; a recent low-contamination irony set is matched only at its own low human ceiling, ruling out memorization as the explanation. A mixed-effects model confirms the mechanism: judge correctness is governed overwhelmingly by item-level human (dis)agreement (random-intercept $SD_{\text{item}} = 2.02$) and by task subjectivity ($OR = 0.17$ per unit of 1-ceiling), while model scale contributes only modestly ($OR = 1.30$ per $\log_{10}$ params); after FDR correction 22 of 36 judge $\times$ dataset cells sit significantly below the human ceiling. Together the studies yield a decision framework telling HCC researchers when an LLM judge is a validated substitute, a human-in-the-loop assistant, or an unsafe replacement.

Additional Key Words and Phrases: LLM-as-a-judge, human-centered computing, systematic review, qualitative coding, annotation, human-AI agreement, evaluation methodology, data contamination

1 INTRODUCTION AND MOTIVATION

In barely two years, LLMs have moved from objects of HCI study to *instruments* of HCI method. Researchers now use them to code interview transcripts [19, 50], run thematic analyses [14, 40], annotate social-media corpora [20, 45], simulate survey respondents and “silicon samples” [2], generate personas [42], and even perform heuristic usability evaluation [22]. The appeal is obvious: human judgment is the costly bottleneck of human-centered research, and an LLM that labels for \$0.003 an item is a tempting stand-in [20].

The danger is equally clear. HCC and qualitative methods are not mere labeling pipelines; they carry *epistemological commitments* (interpretation, reflexivity, positionality, and the situated knowledge of the researcher) that an LLM cannot obviously honor [15, 44]. Substituting a model for a human participant can *flatten* and misportray the very identity groups HCI seeks to center [12, 48]. And the empirical record is genuinely mixed: the same model can match two

Study 1a: *Where is LLM-as-a-judge reliable?* (cross-domain review, 217 studies) → **Study 1b:** *Is it a moving target?* (most powerful open models vs reported numbers, SummEval) → **Study 2:** *How does HCC use LLMs for human-input tasks?* (55 studies) → **Study 3:** *Fill the gaps*, standardized evaluation on HCC datasets with human labels (open models via an institutional vLLM API).

Fig. 1. The four-study progression. Study 1a maps where LLM judges are reliable; Study 1b tests whether today’s strongest open models move that map; Study 2 narrows to Human-Centered Computing practice; Study 3 supplies new empirical evidence on under-tested HCC datasets.

human coders on a structured codebook [17] yet hallucinate usability problems that do not exist [22], or agree with a second annotator on stance [20] yet be fooled into mislabeling relevance by a few injected keywords [1].

HCC researchers therefore face a question with no consolidated answer: **for which HCC research tasks is an LLM-as-a-judge a safe substitute for human judgment, for which is it a useful assistant, and for which is it unsafe?** This paper answers that question through three studies with a deliberate progression (Figure 1):

- (1) **Study 1a, the general landscape.** A PRISMA systematic review of 217 studies across 29 domains establishes *where, in general*, LLM-as-a-judge agrees with humans, anchored to the human-human ceiling. This gives HCC the broad map and the methodological vocabulary (the reliability rubric, the bias catalogue, the decision map).
- (2) **Study 1b, is the landscape a moving target?** The review captures results *as reported*, often with one- or two-generation-old proprietary judges. We re-run a flagship “promising”-band benchmark (SummEval summarization quality) with the *most powerful open models available today* and ask whether they now surpass the canonical reported judge correlations, separating limitations that are a “matter of time” (scale/recency) from *structural* ceilings that more capable models do not cross.
- (3) **Study 2, the HCC lens.** A second PRISMA review of 55 studies zooms into the methods HCC actually uses (qualitative coding, thematic/content analysis, crowdsourced annotation, participant simulation, persona generation, usability evaluation, and design critique) and codes how the field positions LLMs (augment / replace / simulate) and with what epistemic stance.
- (4) **Study 3, filling the gaps.** Studies 1-2 reveal HCC datasets that contain human labels but have never been formally evaluated with an LLM judge. Study 3 runs a standardized LLM-as-a-judge meta-evaluation on these datasets, using open models served through an institutional vLLM API and the agreement-against-human-ceiling methodology distilled from the prior literature, with a mixed-effects analysis of what governs judge reliability.

Contributions. (1) The first consolidated, human-ceiling-anchored map of LLM-as-a-judge reliability across 29 domains (Study 1a). (2) A direct test of whether the most powerful open models available today surpass the reported judge correlations in a flagship “promising” domain, adjudicating “matter of time” vs. structural ceiling (Study 1b). (3) The first systematic review of LLM substitution *specifically* for HCC/HCI/qualitative methods, with a method×role×stance coding that surfaces the field’s augment-not-replace consensus and its epistemic tensions (Study 2). (4) A reproducible, standardized evaluation harness plus a mixed-effects account of what governs judge reliability (Study 3), tested on HCC datasets that had human labels but no prior LLM-judge evaluation, plus two released corpora (217 + 55 coded studies), all code, and a decision framework for practitioners.

2 BACKGROUND AND SHARED METHODOLOGY

LLM-as-a-judge descends from reference-free metrics (BERTScore, COMET) but uses a general instruction-following model with a natural-language rubric, supporting pointwise, pairwise, and listwise judgment [21, 32]. Four architectural families recur: prompted general LLMs [33, 52]; fine-tuned open evaluators [28, 47, 54]; juries/multi-agent panels [11, 46]; and calibrated rubric methods [23]. Prior surveys taxonomize methodology [21, 31]; we instead synthesize *where substitution is empirically validated*, and, uniquely, do so through the HCC lens.

Related work. Four literatures meet in this paper. (1) *LLM-as-a-judge evaluation*: studies report strong agreement on aggregate, verifiable tasks (MT-Bench [52], GEMBA [29], SAFE factuality [49]) but document systematic biases (position, verbosity, self-preference, and an expertise gap by which judges converge on crowd rather than expert consensus [4]), which motivates benchmarking against the human ceiling rather than against a single “gold.” (2) *LLMs in HCI and qualitative research*: a fast-growing body uses LLMs to assist coding and thematic analysis [14, 19, 50] while qualitative scholars warn that interpretation, reflexivity, and positionality resist automation [15, 44]. (3) *Participant and persona simulation*: “silicon samples” [2] and synthetic users are increasingly scrutinized for flattening identity and distorting variance [5, 12, 48]. (4) *Learning with disagreement and perspectivism*: work on annotator disagreement and human label variation [7] argues that for subjective constructs the “ceiling” is itself low and contested, which our Study 3 datasets (ChaosNLI, GoEmotions, MultiPICO) operationalize directly. We connect these threads with two methodological commitments often missing from judge evaluations: anchoring every verdict to the *human-human* ceiling, and probing *data contamination* so that agreement on older public datasets is not mistaken for genuine competence.

Shared protocol. Both reviews follow PRISMA 2020 [38], identify studies through a triangulated search (the OpenAlex API as the primary bibliographic database, Semantic Scholar as a cross-check, and a structured web “deep-research” arm), and code each study on a five-level **reliability verdict** benchmarked against the *human-human agreement ceiling* for the same task: **1 Validated** ($\approx / >$ ceiling), **2 Promising** (conditional), **3 Mixed** (task-dependent), **4 Caution** (below ceiling), **5 Unreliable** (near chance / bias-dominated). We report a reliability score = 6 – verdict and synthesize within metric families rather than pooling incommensurable effect sizes. Full protocols, search logs, and the two coded corpora are released.

3 STUDY 1A: THE CROSS-DOMAIN RELIABILITY LANDSCAPE

3.1 Method (summary)

Study 1a systematically reviewed 217 studies across 29 domains comparing LLM-as-a-judge to human annotation. Identification triangulated OpenAlex (1500 records, 957 unique), Semantic Scholar, and a 59-query web search; screening retained on-topic empirical comparisons; 29 studies reported an extractable numeric agreement value (Figure 2; the full coded corpus is Table 6).

3.2 Findings: a green/amber/red landscape

The modal verdict is *mixed/task-dependent*; only 31% of studies sit in the reliable band and 19% in the problematic band (mean reliability 3.15/5). Reliability is governed by three properties: **objectivity**, **tractable verification**, and a **low human-agreement floor**. **Green (≥ 3.5 ; 8 domains)**: machine translation (system level), text-to-image alignment, verifiable instruction-following, QA/factuality, information extraction, general-NLG pairwise preference, dialogue, and legal scoring: here LLM judges frequently match or exceed the human ceiling (e.g., MT-Bench 85% vs. 81% human-human

Figure 1. PRISMA 2020 flow - two-arm identification (OpenAlex database + web/citation)

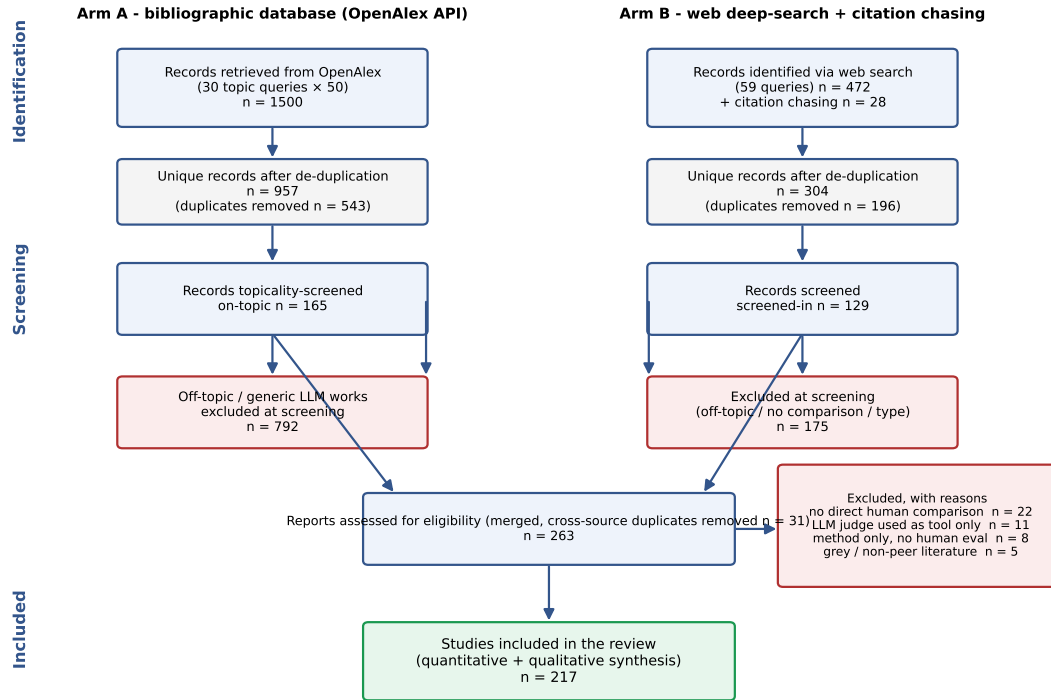


Fig. 2. Study 1a PRISMA 2020 flow (two-arm identification: OpenAlex database + web/citation).

[52]; GEMBA-MQM 96.5% [29]; SAFE wins 76% of disagreements [49]). **Amber (19 domains)**: summarization, RAG, education, social-science annotation, medicine (structure-dependent: $\kappa \approx 0.82$ -0.93 structured vs. 0.47-0.71 complex reasoning), code, and IR/relevance (strong but contested [13]). **Red (2 domains)**: low-resource multilingual evaluation and creative writing (near-zero expert correlation [10]). Figures 3a-3b give the domain ordering and the domain×task landscape; the decision map (Figure 4b) positions domains by reliability against stakes.

3.3 Failure modes and the decision map

Recurring biases such as verbosity/length, position, self-preference, style-over-substance, preference leakage, non-determinism, and the *expertise gap* (judges converge to crowd, not expert, consensus [4]) all modulate reliability (Figure 4a). Agreement should be benchmarked against the human ceiling, which on subjective tasks is itself low (Krippendorff $\alpha < 0.35$ [27]). The Study-1a decision map (Figure 4b) and Table 1 translate this into deployment guidance and motivate both the moving-target test (Study 1b) and the HCC-specific analysis that follows.

4 STUDY 1B: IS THE RELIABILITY LANDSCAPE A MOVING TARGET?

A systematic review necessarily captures results *as reported*, and most “promising”-band evidence in Study 1a was produced with proprietary judges that are now one or two model generations old (e.g. the influential G-Eval used

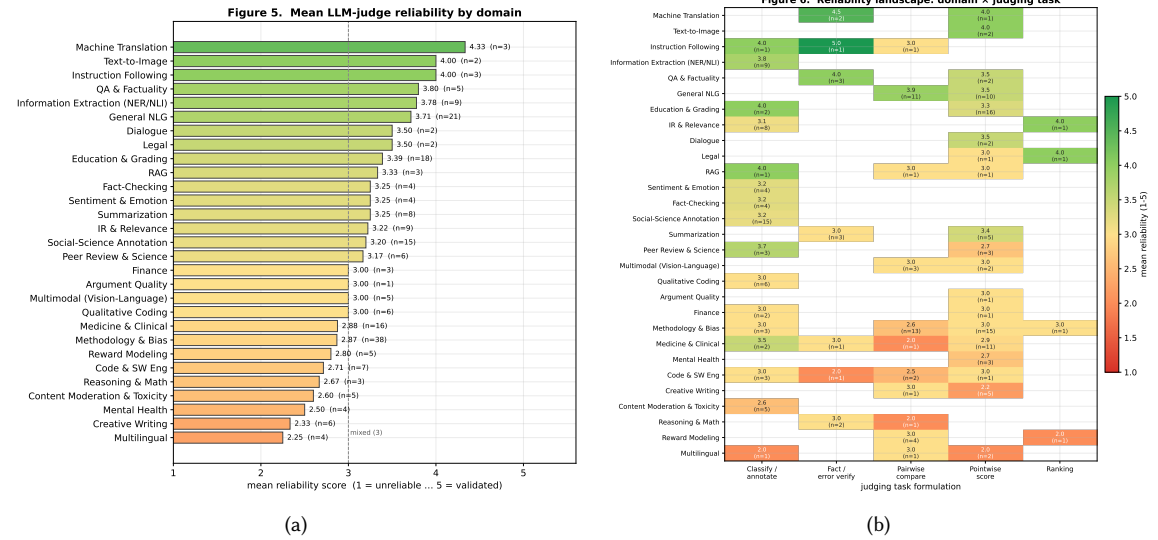


Fig. 3. Study 1a. (a) Mean reliability by domain (green/amber/red). (b) Reliability landscape: domain x judging task; verification/classification columns are greener than open-ended scoring.

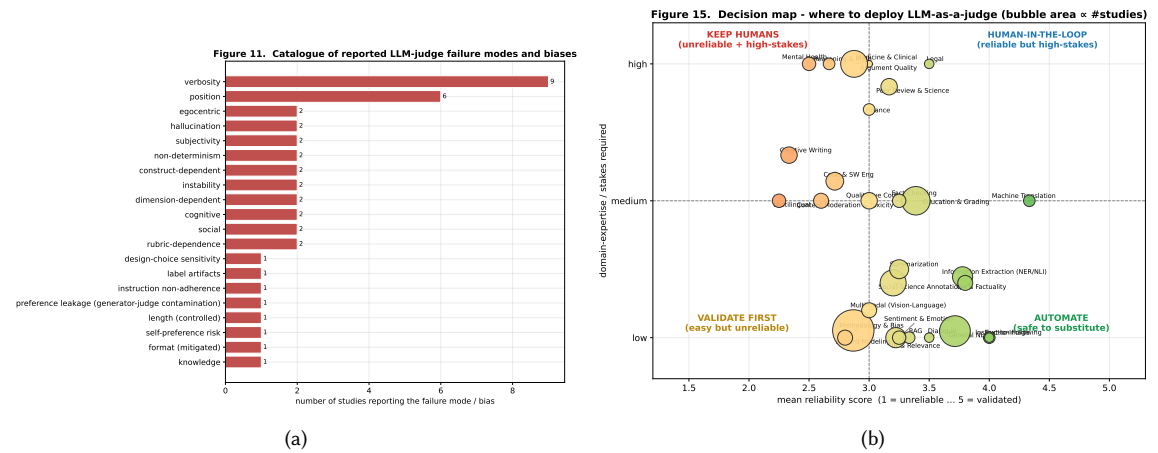


Fig. 4. Study 1a. (a) Catalogue of reported failure modes/biases. (b) Deployment decision map (reliability x stakes; bubble area proportional to number of studies).

GPT-4). This raises a question that the review alone cannot answer: when an LLM judge falls short of the human ceiling in a promising domain, is that a *temporary* limitation that more capable models will erase (“matter of time”), or a *structural* property of LLM-as-a-judge that better models do not overcome? Study 1b answers this directly for a flagship case by re-running the evaluation with the most powerful open models available to us today and comparing against the canonical reported numbers.

Table 1. Domain-level decision table. Mean reliability score (1 = unreliable . . . 5 = validated), modal verdict, and traffic-light signal (GREEN ≥ 3.5 ; AMBER 2.5-3.5; RED < 2.5). Sorted by mean reliability. Small- n domains should be read together with the family-level aggregates.

Domain / task	n	Mean rel.	Modal verdict	Signal
Machine Translation	3	4.33	Promising	GREEN
Text-to-Image	2	4.00	Promising	GREEN
Instruction Following	3	4.00	Validated	GREEN
QA & Factuality	5	3.80	Promising	GREEN
Information Extraction (NER/NLI)	9	3.78	Promising	GREEN
General NLG	21	3.71	Promising	GREEN
Dialogue	2	3.50	Promising	GREEN
Legal	2	3.50	Promising	GREEN
Education & Grading	18	3.39	Mixed	AMBER
RAG	3	3.33	Mixed	AMBER
Summarization	8	3.25	Promising	AMBER
Fact-Checking	4	3.25	Promising	AMBER
Sentiment & Emotion	4	3.25	Mixed	AMBER
IR & Relevance	9	3.22	Promising	AMBER
Social-Science Annotation	15	3.20	Mixed	AMBER
Peer Review & Science	6	3.17	Mixed	AMBER
Multimodal (Vision-Language)	5	3.00	Mixed	AMBER
Qualitative Coding	6	3.00	Mixed	AMBER
Argument Quality	1	3.00	Mixed	AMBER
Finance	3	3.00	Mixed	AMBER
Medicine & Clinical	16	2.88	Mixed	AMBER
Methodology & Bias	38	2.87	Mixed	AMBER
Reward Modeling	5	2.80	Mixed	AMBER
Code & SW Eng	7	2.71	Mixed	AMBER
Reasoning & Math	3	2.67	Mixed	AMBER
Content Moderation & Toxicity	5	2.60	Mixed	AMBER
Mental Health	4	2.50	Mixed	AMBER
Creative Writing	6	2.33	Mixed	RED
Multilingual	4	2.25	Caution	RED

4.1 Method

We select **SummEval** [18] (summarization quality assessment, a central amber-band task in Study 1a) because it pairs (i) a widely cited LLM-judge result to beat (G-Eval/GPT-4; 33) with (ii) public expert human ratings on four aspects (coherence, consistency, fluency, relevance) for $100 \times 16 = 1600$ machine summaries. We replicate the G-Eval summary-level protocol *verbatim* (same aspect definitions, 1-5 integer ratings, temperature 0) but swap the judge for the 4 most powerful open models in the available open-model catalogue (deepseek-v4-pro, glm-5.1, gptoss-120b, qwen3.6-35b-a3b). The comparison metric is the canonical *summary-level Spearman*: for each source document, correlate the judge’s 16 scores with the human means, averaged across documents. The bar to beat is G-Eval/GPT-4’s reported per-aspect correlations (coherence .582, consistency .507, fluency .455, relevance .547; average $\approx .52$). In total Study 1b contributes 25,600 fresh judgments.

4.2 Results

Table 2 reports each open model’s summary-level Spearman against the reported G-Eval bar, and Figure 5 plots them per aspect. The most powerful open models now *match or exceed* the reported proprietary-judge correlations on

Table 2. Study 1b. Summary-level Spearman ρ of the most powerful open judges on SummEval vs. the reported G-Eval-4/GPT-4 correlations (25,600 judgments). Green cells *surpass* the reported per-aspect number.

Judge	Coher.	Consist.	Fluency	Relev.	Avg ρ
<i>Reported (G-Eval-4)</i>	0.582	0.507	0.455	0.547	0.523
deepseek-v4-pro	0.548	0.688	0.392	0.540	0.542
qwen3.6-35b-a3b	0.606	0.702	0.298	0.540	0.536
glm-5.1	0.650	0.693	0.217	0.543	0.526
gptoss-120b	0.586	0.685	0.260	0.482	0.503

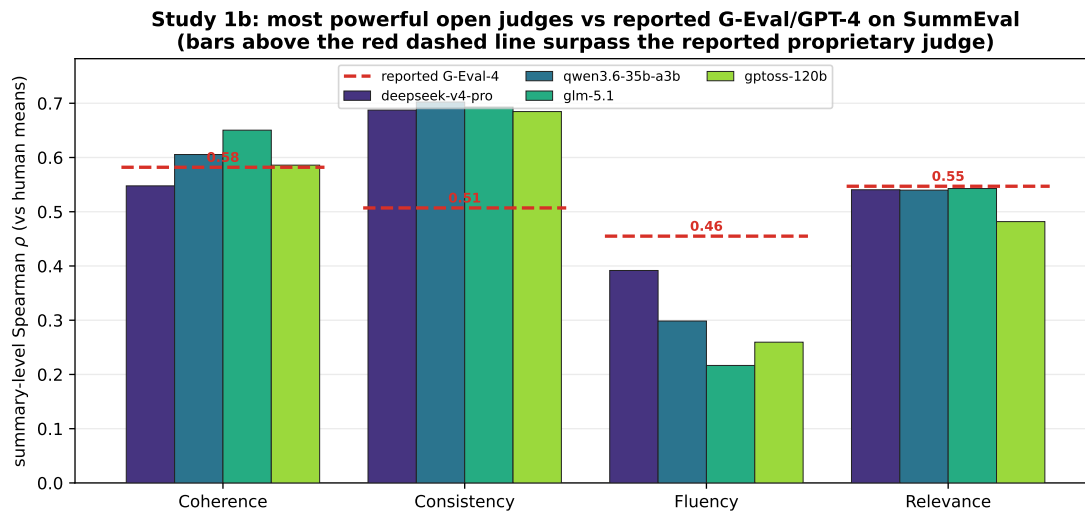


Fig. 5. Study 1b. Summary-level Spearman of the most powerful open judges on SummEval, per aspect, against the reported G-Eval/GPT-4 bar (dashed). Bars above the dashed line indicate an open model that *surpasses* the reported proprietary-judge correlation.

SummEval. 3 of the 4 open judges beat G-Eval’s reported *average* correlation, led by deepseek-v4-pro (avg summary-level $\rho = 0.542$ vs. the reported ≈ 0.52). The aspect pattern is itself the finding. On **consistency** (the most *verifiable* judgment, groundable in the source document) *all four* models leap past the reported value (from .507 to .685-.702), and on **coherence** three of four do (up to .650 vs. .582); **relevance** is on par (.48-.54 vs. .55). But on **fluency** every open model falls well *short* (.22-.39 vs. a reported .455): human fluency ratings of competent machine summaries are compressed near the top of the scale (low variance), so the summary-level correlation is structurally low *regardless of how capable the judge is*, a limit that lives in the human reference, not the model, and that scale does not move.

Interpretation: time vs. structure. Study 1b shows the reliability landscape is *partly* a moving target. Where Study 1a’s amber rating reflected a capability gap on a verifiable sub-judgment, today’s open models close or exceed it; reliability there is largely a matter of time and access. But where the shortfall reflects a *compressed or contested human ceiling* (e.g. fluency, and, as Study 3 will show, subjective social annotation), even the most powerful models do not break through, because the limit lives in the human reference, not the model. This distinction, temporal capability gap vs. structural ceiling, is the lens we carry into HCC.

Takeaway for HCC. Study 1a says LLM judges are safe where the task is objective and verifiable, and Study 1b shows that frontier open models are now strong enough to match proprietary judges on exactly those verifiable tasks. But HCC’s signature methods (qualitative interpretation, participant study, design judgment) are precisely the *subjective, expert, interpretive* tasks Study 1a flags as risky and Study 1b predicts scale will *not* fix. Study 2 tests whether that prediction holds in the HCC literature itself.

5 STUDY 2: LLM SUBSTITUTION IN HCC, HCI, AND QUALITATIVE METHODS

5.1 Motivation and research questions

Study 1’s map is domain-general; HCC needs a method-specific account. Human-centered research is built on methods that historically *require* human input: qualitative coding, thematic and content analysis, interviews and open-ended survey analysis, crowdsourced annotation, participant studies, persona development, usability/heuristic evaluation, and design critique. Study 2 asks: **(RQ2.1)** which of these methods have been attempted with LLMs, and how reliably do LLM outputs agree with human input? **(RQ2.2)** How does the field position the LLM, to *augment, replace, or simulate* the human? **(RQ2.3)** What epistemic stance (enabling, tension, critical) does the literature take, and what does that imply for HCC practice?

5.2 Method

Following the shared protocol, we ran a Study-2-specific identification arm: 25 HCC/HCI/qualitative concept-block queries against OpenAlex (857 unique works), triangulated with the web arm and with the venue-scoped sweep of Study 1 (CHI, CSCW, UIST, IUI, TOCHI). After topicality screening and eligibility assessment we included 55 studies (Figure 6). We extracted, per study: the HCC **method**; the LLM **role** (replace / augment / simulate); reported agreement and the human baseline; the reliability **verdict**; the **epistemic stance** (enabling / tension / critical); and the **community** (HCI, CSCW, qualitative-methods, computational social science, NLP, health). The full corpus is Table 7.

5.3 RQ2.1: Which methods, and how reliable?

LLMs have been compared against human input across 15 distinct HCC methods (Figure 7a), most heavily in *thematic analysis, synthetic participants, and crowdsourced annotation*. Reliability is strongly method-dependent (Figure 7b; Table 3). The *same objectivity/interpretation axis* from Study 1 governs HCC: **objective crowd-annotation** is the most reliable method to substitute: ChatGPT matches or beats crowd workers and even trained coders on stance, topic, and frame labeling [20, 45]; **deductive/inductive coding and content/thematic analysis** are reliable *as augmentation*, with human-equivalent agreement on structured codebooks (GPT-4 exceeding substantial κ on 8/9 hermeneutic tasks [17]; Gemini $\kappa = 0.907$ in multi-LLM thematic analysis [26]; 79.7% agreement in health thematic summarization [9]) but only moderate agreement in open interpretive settings ($\kappa \approx 0.40\text{--}0.46$ [30, 51]); and **participant/persona simulation and usability evaluation** are the least reliable.

5.4 RQ2.2: Augment, replace, or simulate?

HCC overwhelmingly positions LLMs to *augment* humans, not replace them: 36 of 55 studies frame the LLM as a collaborator/assistant, versus 14 as a replacement and 5 as a human simulator (Figure 8a). This is not merely rhetorical caution; it tracks reliability (Figure 9): augmentation studies skew reliable, whereas *replacement* is validated only for objective annotation and *simulation* of participants is the least reliable role (mean reliability augment 3.17 vs. replace

Figure S2-1. Study 2 identification (HCC/HCI/qualitative methods)

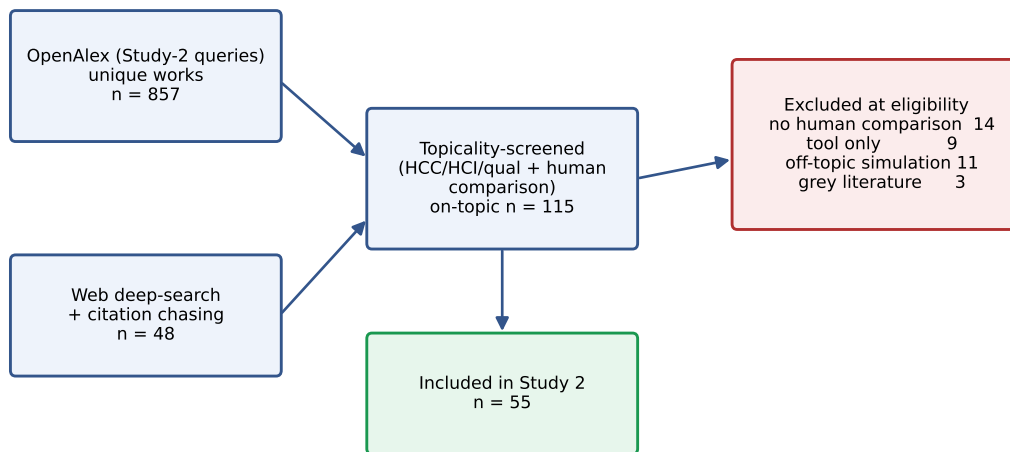


Fig. 6. Study 2 identification flow (HCC/HCI/qualitative methods).

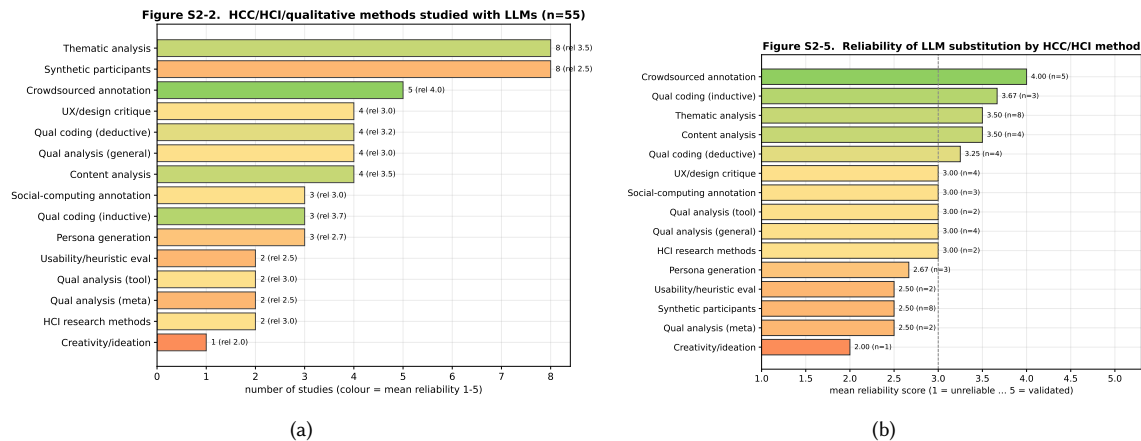


Fig. 7. Study 2. (a) HCC/HCI/qualitative methods studied with LLMs (colour = mean reliability). (b) Mean reliability of LLM substitution by method.

3.07 vs. simulate 2.8). The replacement mean is buoyed by objective crowd-annotation; remove it and replacement of *interpretive* human input is firmly amber/red. The canonical HCC workflow that emerges is human-in-the-loop: LLM proposes, human vets and retains interpretive authority [14, 19, 35].

5.5 RQ2.3: Epistemic stance and the limits of substitution

The dominant stance is *tension* (38/55), with 8 explicitly *critical* and only 9 unreservedly *enabling* (Figure 8b). Three threads recur. (i) **Interpretation and reflexivity**. Qualitative scholars warn that LLMs perform surface-level extraction well but cannot supply the situated, reflexive interpretation that gives qualitative findings their validity, and that

Table 3. Study 2 method-level decision table: study count, modal LLM role, mean reliability, and traffic-light signal (GREEN ≥ 3.5 ; AMBER 2.5-3.5; RED < 2.5).

Method	<i>n</i>	Modal role	Mean rel.	Signal
Crowdsourced annotation	5	replace	4.00	GREEN
Qual coding (inductive)	3	augment	3.67	GREEN
Content analysis	4	augment	3.50	GREEN
Thematic analysis	8	augment	3.50	GREEN
Qual coding (deductive)	4	augment	3.25	AMBER
HCI research methods	2	augment	3.00	AMBER
Qual analysis (general)	4	augment	3.00	AMBER
Qual analysis (tool)	2	augment	3.00	AMBER
Social-computing annotation	3	replace	3.00	AMBER
UX/design critique	4	augment	3.00	AMBER
Persona generation	3	augment	2.67	AMBER
Qual analysis (meta)	2	augment	2.50	AMBER
Synthetic participants	8	simulate	2.50	AMBER
Usability/heuristic eval	2	replace	2.50	AMBER
Creativity/ideation	1	augment	2.00	RED

Figure S2-3. How HCC research positions LLMs: augment vs replace, and epistemic stance

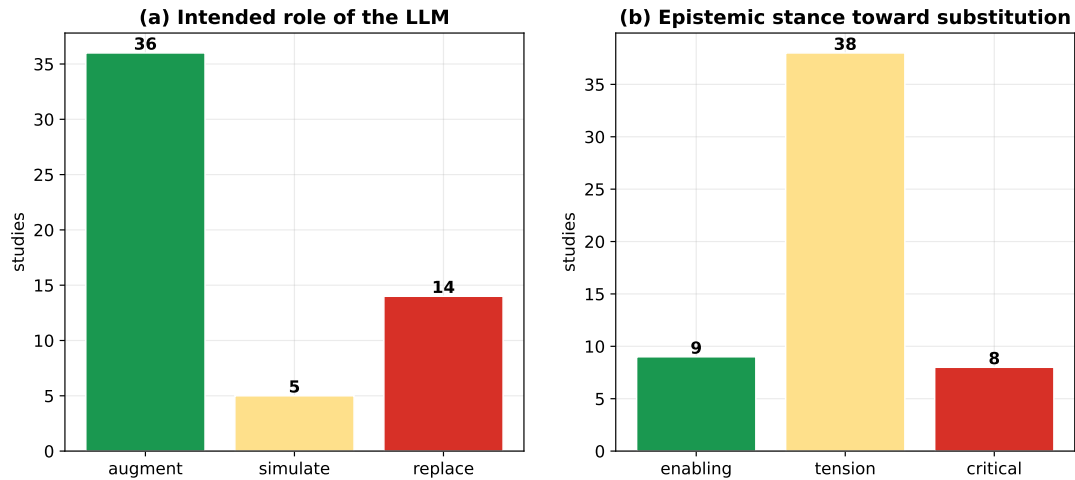


Fig. 8. Study 2. (a) Intended LLM role across the HCC corpus; augmentation dominates. (b) Epistemic stance toward substitution; *tension* dominates.

“conditional trust” in tool output risks hollowing out analytic rigor [15, 36, 37, 44]. (ii) **Participants and identity.** Replacing human participants with LLMs *flattens and misportrays* demographic groups, positionality cannot be simulated [48], and LLM “subgroups” are often caricatures [12]; synthetic survey data distort variances and correlations [5], and an LLM is worth only a bounded number of real respondents [25]. HCI critiques accordingly recast synthetic users as “objects of inquiry,” not stand-ins [24, 43]. (iii) **Expert design judgment.** GPT-4o overlaps only 21% with expert-found usability issues and hallucinates false positives [22], and LLM UI critique trails expert designers [16], though synthetic

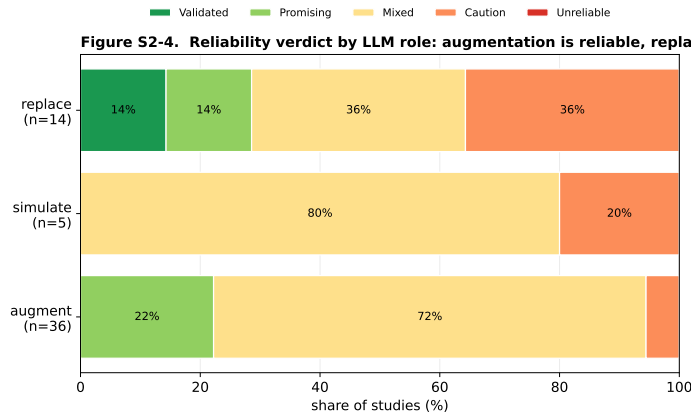


Fig. 9. Study 2. Reliability verdict by LLM role. Augmentation is reliable; replacement is reliable only for objective annotation; participant/usability *simulation* is the least reliable.

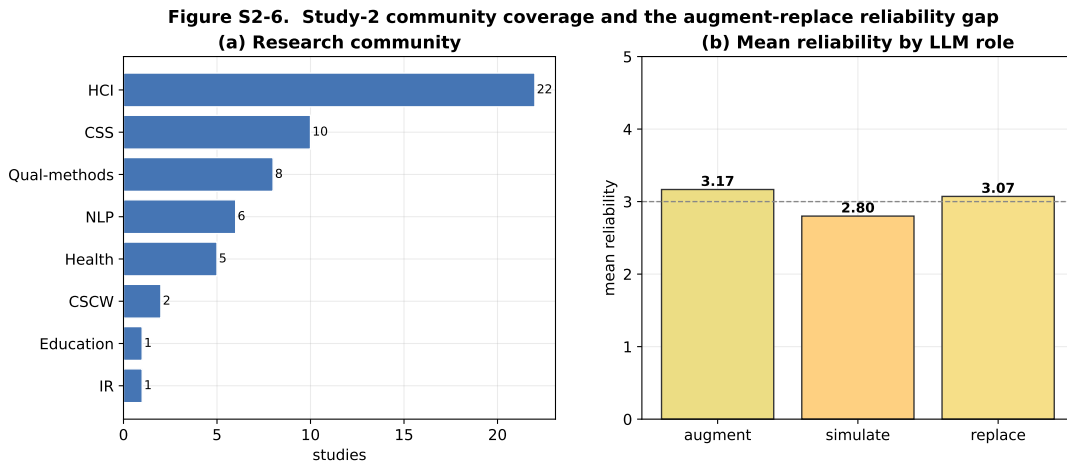


Fig. 10. Study 2. (a) Coverage by research community. (b) Mean reliability by LLM role, augmentation > replacement > simulation.

heuristic evaluation can exceed *novice* evaluators on issue recall [53]. Figure 10 shows coverage skews to HCI and computational social science, with the augment-replace reliability gap visible across communities.

5.6 Study 2 synthesis

For HCC research, the evidence supports a clear, conditional posture. **Validated to substitute:** objective crowd/data annotation (stance, topic, relevance, frames) where a human ceiling is modest and verifiable. **Reliable only as augmentation (human-in-the-loop):** deductive/inductive qualitative coding and thematic/ content analysis: LLMs accelerate breadth and first-pass coding, but humans must own interpretation, theme development, and reflexive accounting. **Unsafe to substitute:** simulating human participants, generating personas as stand-ins for real users, and replacing expert usability/design judgment: these flatten identity, fabricate issues, and forfeit the situated knowledge

that defines human-centered inquiry. This posture both confirms Study 1’s objective/subjective prediction *within* HCC and exposes concrete gaps (§6) where HCC datasets with human labels have never been formally tested with an LLM judge.

6 STUDY 3: AN EMPIRICAL LLM-AS-A-JUDGE EVALUATION ON HCC DATASETS WITH HUMAN GROUND TRUTH

Studies 1-2 expose a recurring gap: many human-centered datasets ship with human labels yet have never had a *formal* LLM-as-a-judge evaluation, meaning a panel of models, bias controls, agreement benchmarked against the human ceiling, and a contamination check. Study 3 closes this gap empirically with a released, reproducible harness.

6.1 Method

We evaluate a panel of 6 open models served via an **institutional vLLM API** (OpenAI-compatible), spanning five families and 9B-754B parameters (g1m-5.1 754B, gptoss-120b, deepseek-v4, gemma-4-31b, qwen3.6-27b, qwen3.5-9b), run in parallel at each model’s documented concurrency. Each model judges every item as a *virtual annotator* under the same instructions the humans received. We benchmark against the *human-human ceiling* (random-annotator-vs-majority accuracy and pairwise agreement, computed from the per-item annotator counts), report accuracy, Cohen’s κ , and macro-F1 against the majority gold, compute a count-based **alt-test** winning rate [7], aggregate the panel into a majority **jury** [46], and run a verbatim quote-completion **contamination probe**. Sample size is power-justified at $n=500$ items/dataset (95% CI half-width $\leq \pm 4.4\%$; $>80\%$ power for a 0.10 gap vs the ceiling). The run comprised **18000 judgments** across 6 datasets with *zero* API errors and a 100% answer-parse rate; only item-level labels and aggregates are released (no raw text), and the two sensitive sets are used under their public licenses.

6.2 Results

Figures 11-12 and Table 4 yield one organizing finding: **the human-agreement ceiling governs LLM reliability**. Where humans agree little (NLI; ceiling 0.64-0.75) the panel *matches or exceeds* them: on ChaosNLI-SNLI 5/6 models are **Validated** (jury 0.858 $>$ ceiling 0.752), and ChaosNLI-MNLI is **Promising**. Where humans agree strongly on a subjective social construct, the panel falls short: GoEmotions 4-way sentiment (ceiling 0.78) is uniformly **Mixed** (best 0.61); HateXplain 3-way hate (ceiling 0.835) is **Mixed** at best and **Unreliable** for the small model; binary offensiveness (SBIC, ceiling 0.905) is the strongest social-computing case (**Promising**; g1m-5.1 0.856) yet still below the ceiling. The sixth dataset is a deliberate *clean condition*: MultiPICO irony detection [8], a 2024 perspectivist corpus recent enough to be unlikely in pretraining. Its human ceiling is itself low (0.82, just above the 76% majority baseline); the panel *matches* it (**Promising**; best 0.82, jury 0.83) but largely by defaulting to the majority class (macro-F1 ≈ 0.45), and qwen3.5-9b collapses (0.40, **Unreliable**). Four further effects: (i) **granularity**: binary offensiveness (Promising) clearly beats 3-way hate (Mixed); (ii) **scale**: qwen3.5-9b is weakest on *every* dataset (Caution/Unreliable), while 27B-754B cluster together (bigger is not always better: 31B gemma ties the 754B g1m); (iii) the **alt-test** (winning rate ≥ 0.5) is passed only on the *low-ceiling* tasks, NLI and the binary irony set (5/6 models on both ChaosNLI-SNLI and MultiPICO); on every *high-consensus* task, including the Promising binary offensiveness (SBIC, 0/6), no model earns the statistical right to replace a human annotator; (iv) **juries** raise accuracy but do not lift any subjective task above its human ceiling. The **contamination probe** (Table 4, last column) shows the strongest judges have non-trivial verbatim recall across sets (gemma MEDIUM throughout), caveating the NLI “Validated” result; crucially the recent low-contamination MultiPICO

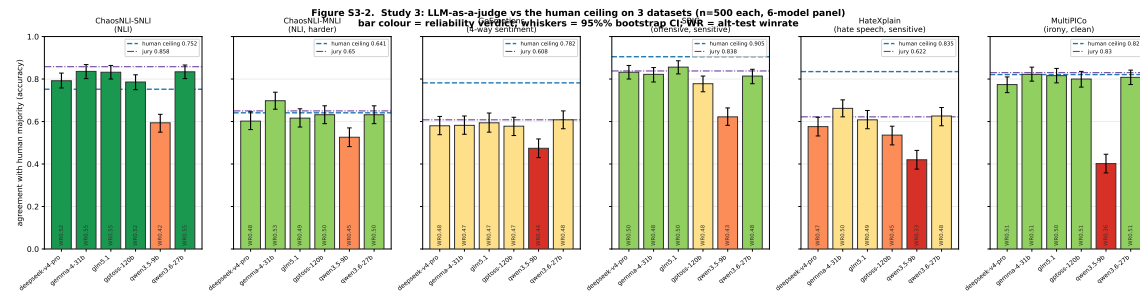


Fig. 11. Study 3. Per-dataset accuracy vs the human ceiling (blue dashed) for the 6-model panel (n=500 each; whiskers = 95% bootstrap CI; bar colour = verdict; WR = alt-test winning rate). Datasets ordered by construct subjectivity. LLMs clear LOW ceilings (NLI) but fall short of HIGH ceilings on subjective social-computing tasks.

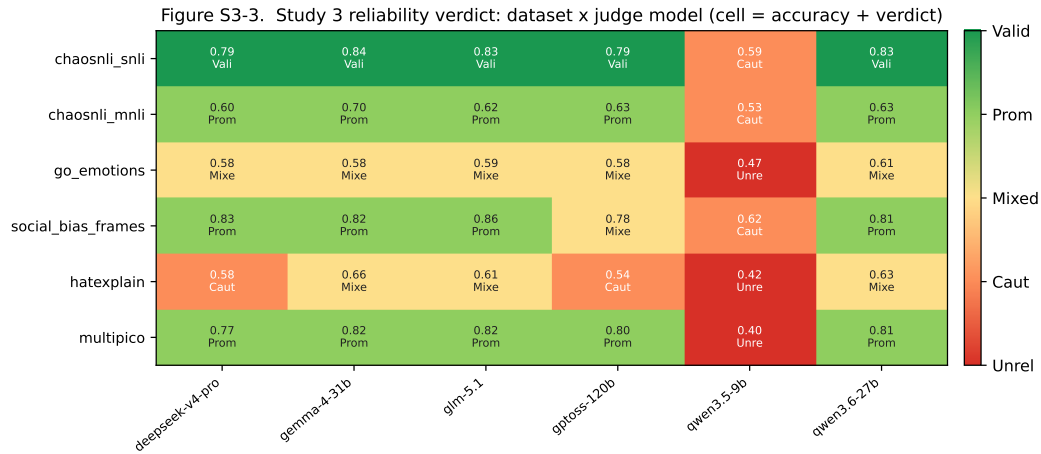


Fig. 12. Study 3. Dataset×judge-model reliability verdict (cell = accuracy + verdict). The scale effect (qwen3.5-9b column) and the construct-subjectivity gradient (rows) are both visible.

Table 4. Study 3 results across 6 datasets × 6 models (18000 judgments). Human ceiling = random-annotator-vs-majority accuracy; alt-test pass = winning rate ≥ 0.5; contamination tier from the verbatim-recall probe.

Dataset	Construct	Human ceiling	Best model (acc)	Jury (acc)	Modal verdict	alt-test pass	Contam. tier
ChaosNLI-SNLI	NLI	0.75	0.84 (gemma-4-31b)	0.86	Validated	5/6	MEDIUM
ChaosNLI-MNLI	NLI	0.64	0.70 (gemma-4-31b)	0.65	Promising	1/6	MEDIUM
GoEmotions	emotion (4-way)	0.78	0.61 (qwen3.6-27b)	0.61	Mixed	0/6	MEDIUM
SBIC	offensive (binary)	0.91	0.86 (glm-5.1)	0.84	Promising	0/6	MEDIUM
HateXplain	hate (3-way)	0.83	0.66 (gemma-4-31b)	0.62	Mixed	0/6	MEDIUM
MultiPiCo (clean)	irony (binary)	0.82	0.82 (gemma-4-31b)	0.83	Promising	5/6	MEDIUM

(LOW for the small-recall qwen3.5-9b) is still matched only at its low human ceiling, so the social-computing shortfall tracks the human ceiling, not memorization.

Figure S3-4. Study 3 inferential layer: what governs LLM-judge agreement

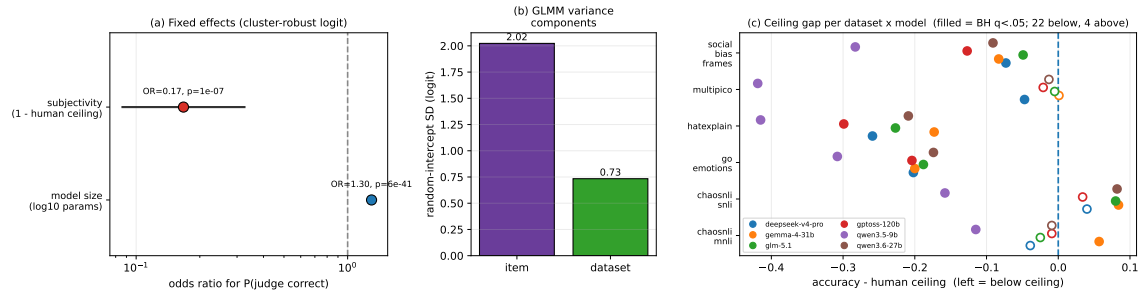


Fig. 13. Study 3 inferential layer. (a) Fixed-effect odds ratios (item-cluster-robust logit): subjectivity is a strong negative predictor of judge correctness, scale a weak positive one. (b) GLMM variance components: item-level variance dominates. (c) Ceiling gap per judge×dataset (filled = BH $q < .05$; 22 below, 4 above).

6.3 What governs reliability? A mixed-effects analysis

To test the “ceiling governs” claim inferentially rather than descriptively, we model every individual judgment ($N=18000$; outcome = 1[judge = majority gold]) with a logistic specification whose fixed effects are task *subjectivity* (1-human-ceiling accuracy, at the dataset level) and model *scale* ($\log_{10}$ parameters), reported both as a Bayesian mixed-effects logistic GLMM with random intercepts for dataset and item, and, for clean inference, as a logistic GLM with item-cluster-robust standard errors (Figure 13). Three results stand out. (i) **Subjectivity dominates**: each unit of 1-ceiling multiplies the odds of agreeing with humans by $OR = 0.17$ ($p = 1e - 7$). (ii) **Scale helps, but modestly**: larger models do better ($OR = 1.30$ per $\log_{10}$ params, $z = 13.40$), an effect far smaller than subjectivity’s, consistent with Study 1b’s “partly a moving target.” (iii) **Variance lives in the items**: the GLMM random-intercept standard deviation is far larger across *items* ($SD = 2.02$) than across datasets ($SD = 0.73$), i.e. *which item* is being judged, its latent human (dis)agreement, explains judge correctness better than which model judges or even which dataset it comes from. Finally, a Benjamini-Hochberg analysis across all 36 judge×dataset cells finds 26 significant at $q < .05$: **22 sit significantly below the human ceiling** and only 4 significantly above (the objective-NLI cases). Scaling the judge shifts cells modestly rightward; it does not move a high-ceiling subjective task across the line.

6.4 Integration with the decision framework

Mapped onto the five-level rubric, Study 3 drops fresh, dataset-level points into the Study-1/2 landscape and *quantitatively confirms* the augment-not-replace conclusion for human-centered annotation: LLM judges are a validated substitute for objective/low-ceiling labeling and a strong assistant for coarse (binary) moderation, but fall short of, and fail the replacement test for, subjective, fine-grained, or high-consensus human-centered constructs, a pattern the mixed-effects model shows is driven by item-level human (dis)agreement rather than by model scale. The recent MultiPICO clean condition rules out contamination as the explanation: a 2024 set the low-recall judge has not memorized is still matched only at its (low) human ceiling. *Limitation*: the panel is limited to open models from a single provider, and qualitative-coding corpora with public multi-coder labels (still scarce) are the immediate next addition.

7 DISCUSSION

7.1 From four research questions to one principle

Our studies were designed as a narrowing sequence, and their research questions answer one another. **Study 1a** asked *where*, across domains, an LLM-as-a-judge agrees with humans; its goal was to give HCC a map and a vocabulary, and it found that reliability tracks three properties (objectivity, tractable verification, and a low human-agreement floor). **Study 1b** asked whether that map is a *moving target*; its goal was to separate temporal from structural limits, and it found that today’s most powerful open models already match or exceed proprietary judges on the *verifiable* part of a promising task (consistency on SummEval) but not on the part whose human ceiling is compressed (fluency), so capability closes some gaps and cannot close others. **Study 2** asked, of the methods HCC actually uses, which have been attempted with LLMs and how reliably (RQ2.1), how the field positions the model as augment/replace/simulate (RQ2.2), and with what epistemic stance (RQ2.3); its goal was the HCC-specific account, and it found an augment-not-replace consensus whose reliability follows the same objective/subjective axis as Study 1a, under a dominant stance of *tension*. **Study 3** asked whether that prediction holds when measured directly on HCC datasets with human labels, and *what governs* the outcome; its goal was to fill the empirical gaps and identify the mechanism, and it found that the human-agreement ceiling governs reliability, that item-level disagreement (not model scale, and not contamination) is the dominant source of variance, and that even the strongest panel fails the alt-test wherever humans agree strongly.

These answers converge on a single principle that we offer as the paper’s organizing contribution:

An LLM-as-a-judge can be reliable only to the degree that the human task has a high, verifiable agreement ceiling; it cannot exceed an agreement level that human annotators themselves do not reach, and scaling the model does not raise that ceiling.

This reframes the field’s animating question. “Can a machine judge like a human?” is under-specified; the answerable question is *whether there is a stable, shared human answer for the judge to approximate*. Where there is (objective labeling, verifiable sub-judgments), frontier models already clear the bar; where there is not (interpretation, identity, contested constructs), no amount of capability manufactures a ground truth that the human community does not itself possess.

7.2 Connections to HCI and measurement scholarship

Our results give empirical weight to three long-running conversations in HCI and adjacent fields. **(1) The myth of a single ground truth.** CSCW and HCI have argued that inter-rater reliability is a negotiated practice, not a natural fact [34], and NLP has reframed annotator disagreement as signal rather than noise [3, 41]. Study 3 operationalizes this: when we benchmark judges against the human-human ceiling rather than a manufactured “gold,” the ostensibly poor performance on emotion, hate, and irony is revealed as a property of the *construct* (low or contested human consensus), not a fixable model deficiency. A judge that reports high accuracy against a forced majority is, on these tasks, reporting agreement with a coin the humans themselves are still flipping. **(2) Interpretation is not labeling.** Reflexive thematic analysis treats coding as situated interpretation, not classification [6], and qualitative HCI scholars warn that “conditional trust” in LLM output hollows out analytic rigor [15, 44]. Study 2’s augment-not-replace consensus and Study 3’s subjective-task shortfall are two views of the same boundary: LLMs accelerate the *mechanical* surface of qualitative work while the interpretive core remains human. **(3) Participants are not simulable.** The “silicon sample” program [2] meets a growing critique that simulated participants flatten identity and distort variance [5, 12, 43, 48]; our MultiPICO result, where models default to the majority reading and erase the minority perspective that perspectivist corpora were built to preserve, is a concrete instance of that flattening at the level of a single subjective label.

7.3 A critical assessment: what these practices do to the field

We read the rush to LLM-as-a-judge in HCC as carrying five structural risks that the community should name. **Illusory validation.** Because agreement on older public benchmarks can reflect memorization rather than competence, “validated” verdicts are not trustworthy without a contamination check; our probe shows non-trivial verbatim recall even of recent data, and a field that omits this step risks certifying judges that merely remember. **Methodological monoculture and deskilling.** Cheap labeling invites the substitution of LLM judgment for the human annotation and reflexive interpretation that train the next generation of HCC researchers; combined with the documented convergence of judges toward crowd rather than expert consensus [4], this risks a slow regression of the field’s interpretive capacity toward the mean of its own training data. **Epistemic injustice and flattening.** Where models default to majority readings of subjective content, the perspectives of minority annotators (often the populations HCC seeks to center) are systematically discounted, a representational harm that a single accuracy number conceals. **False closure on contested constructs.** Reporting one accuracy against a majority gold imposes consensus where the data shows disagreement; our irony case (0.82 accuracy but macro-F1 near 0.45) shows how a high headline number can mask the fact that the model has learned the base rate, not the construct. **Incentive capture.** The very convenience of LLM evaluation pressures authors, reviewers, and venues to treat it as a default, embedding its biases into the published record. Against these we weigh genuine benefits, scale, triage, breadth, reproducibility, first-pass coding, and disagreement surfacing, and conclude that LLM judges should be governed as *instruments requiring calibration and disclosure*, not adopted as oracles.

7.4 Implications for designing experiments that use LLM-as-a-judge

The same evidence yields concrete, reusable guidance. We recommend that any study using an LLM-as-a-judge:

- (1) **Anchor to the human-human ceiling, not a single gold.** Report the ceiling (e.g. random-annotator-vs- majority and pairwise agreement) and the judge-vs-ceiling gap; “near-human” is meaningless without it.
- (2) **Gate replacement claims on a replacement test.** Report the alt-test [7] (or an equivalent leave-one-annotator-out test) before asserting that an LLM can stand in for a human annotator.
- (3) **Probe contamination** for any public or pre-cutoff dataset, and discount validation under high recall; state model training-cutoff relative to dataset release.
- (4) **Match the metric to the construct.** On imbalanced or minority-sensitive tasks report macro-F1 and per-class scores, and prefer disagreement-aware evaluation against the full label distribution (soft labels) over forced-majority accuracy.
- (5) **Control known biases and pin the setup.** Counterbalance position, control length, check self-preference, and fix model identifier, version, decoding (temperature, seed) so the judge is reproducible.
- (6) **Power the comparison.** Pre-specify the effect size of interest, report confidence intervals, and size n accordingly; our analysis shows variance lives at the item level, so also *stratify by item difficulty* and report where the judge fails rather than only its mean.
- (7) **Use juries for aggregate decisions, but not as a ceiling-breaker.** Panels improve stability yet do not lift a subjective task above its human ceiling.
- (8) **Pre-register and release.** Register the construct, the human baseline, and the decision threshold; and release item-level labels and aggregates so others can re-anchor and re-code.

Adopted together, these move LLM-as-a-judge from an ad hoc convenience to a measured instrument with a known operating range, which is the precondition for the field to use it without quietly degrading its own standards of evidence.

7.5 A decision framework for HCC

Mapping the principle onto practice yields three deployment regimes:

- **Automate (with light audit):** objective annotation/labeling of user-generated content such as stance, topic, relevance, frames, and simple sentiment [20, 45], where the ceiling is high and verifiable.
- **Augment, human-in-the-loop:** qualitative coding and thematic/content analysis: use the LLM for first-pass coding, breadth, and disagreement surfacing, but keep human interpretive authority, calibrate on a labeled subset, and report reliability against human coders [19, 35, 39].
- **Keep humans (do not substitute):** simulating participants or replacing them with personas, and replacing expert usability/design judgment: these flatten identity, fabricate evidence, and abandon situated knowledge [22, 43, 48].

8 LIMITATIONS

Both reviews used a single-reviewer, LLM-assisted pipeline (logs released) rather than dual human reviewers; metrics are heterogeneous and synthesized within families rather than pooled; several methods/domains rest on small n (flagged via family/role aggregates); database recall depends on indexing and query phrasing (mitigated, not removed, by three-source triangulation); and findings reflect models available through mid-2026. The empirical studies are bounded too: Study 1b uses one flagship promising-domain benchmark (SummEval), and Study 3’s panel is limited to open models from a single provider and to 6 datasets (one of which, MultiPICO, is included as a recent low-contamination clean condition), with qualitative-coding corpora still in progress. The verdict rubric, though explicit and released, embeds judgement; we therefore release both corpora for re-coding. Full protocol, rubric anchors, dataset cards, verbatim judging prompts, and metric definitions are given in Appendices A to C.

9 ETHICS AND REPRODUCIBILITY

Ethics. Studies 1a and 2 analyze published research, not human subjects. Study 3 reuses public datasets under their licenses; two of them (HateXplain, SBIC) contain hateful or offensive text, which we process only to measure moderation reliability and never reproduce. To avoid redistributing potentially harmful or personally-identifiable content, we release only item-level labels and aggregate statistics, never the raw texts. We report a contamination probe precisely so that “validated” verdicts are not silently inflated by pretraining exposure. The work is dual-use only in the benign sense that better-calibrated LLM judges reduce, rather than increase, the risk of unaccountable automated moderation; our central finding (that LLM judges should not replace humans on subjective, high-consensus social annotation) is itself a guard against premature deployment. **Reproducibility.** Every number in this paper regenerates from released code: the OpenAlex, Semantic Scholar, and web search logs; the two coded corpora; the figure and table generators; the Study-1b SummEval harness and the Study-3 judging/scoring/contamination/mixed-effects harness, all keyed to documented model identifiers served through an institutional vLLM API at temperature 0 with idempotent, resumable checkpoints. The auto-generated statistics macros mean the manuscript cannot drift from the data. Repository: a public repository (URL omitted for anonymous review).

10 CONCLUSION

LLMs are already remaking how Human-Centered Computing research is done. Our four studies show that this is neither to be embraced nor refused wholesale. Across 217 cross-domain studies and 55 HCC-specific studies, the evidence converges on a conditional verdict: an LLM-as-a-judge is a *validated substitute* for objective annotation, a *valuable human-in-the-loop assistant* for qualitative coding and analysis, and an *unsafe replacement* for human participants, personas, and expert design judgment. Study 1b shows that the most powerful open models now match or exceed proprietary judges on the verifiable parts of “promising” tasks, and Study 3 turns the remaining “mixed” verdicts into dataset-level evidence using open, locally-servable models and the field’s own reliability standards. The right question for HCC is never “can a machine judge?” but “*can this machine replace a second human, on this method, at this granularity, under these controls?*”, and this paper makes that question answerable, method by method.

REPRODUCIBILITY

Two coded corpora (Study 1: 217 studies; Study 2: 55 studies), three search logs, PRISMA counts, all figure/table-generation code, the auto-generated statistics, and the Study-3 evaluation harness are released. Everything regenerates via make all.

REFERENCES

- [1] Alaofi et al. 2024. LLMs can be Fooled into Labelling a Document as Relevant. In *SIGIR-AP*. <https://doi.org/10.1145/3673791.3698431>
- [2] Argyle et al. 2023. Out of One, Many: Using Language Models to Simulate Human Samples. *arXiv preprint arXiv:2209.06899* (2023). arXiv:2209.06899
- [3] Lora Aroyo and Chris Welty. 2015. Truth Is a Lie: Crowd Truth and the Seven Myths of Human Annotation. In *AI Magazine*, Vol. 36. 15–24.
- [4] Bavaresco et al. 2024. LLMs instead of Human Judges? A Large Scale Empirical Study across 20 NLP Evaluation Tasks. *arXiv preprint arXiv:2406.18403* (2024). arXiv:2406.18403
- [5] Bisbee et al. 2023. Synthetic Replacements for Human Survey Data? The Perils of Large Language Models. *SocArXiv* (2023). <https://doi.org/10.31235/osf.io/5ecfa>
- [6] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101.
- [7] Calderon et al. 2025. The Alternative Annotator Test for LLM-as-a-Judge: How to Statistically Justify Replacing Human Annotators with LLMs. *arXiv preprint arXiv:2501.10970* (2025). arXiv:2501.10970
- [8] Silvia Casola, Simona Frenda, Soda Marem Lo, Erhan Sezerer, Antonio Uva, Valerio Basile, Cristina Bosco, Alessandro Pedrani, Chiara Rubagotti, Viviana Patti, and Davide Bernardi. 2024. MultiPiCo: Multilingual Perspectivist Irony Corpus. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. 16008–16021.
- [9] Castellanos et al. 2025. Large Language Models for Thematic Summarization in Qualitative Health Care Research: Comparative Analysis of Model and Human Performance. *JMIR AI* (2025). <https://doi.org/10.2196/64447>
- [10] Chakrabarty et al. 2024. Art or Artifice? Large Language Models and the False Promise of Creativity. *arXiv preprint arXiv:2309.14556* (2024). arXiv:2309.14556
- [11] Chan et al. 2023. ChatEval: Towards Better LLM-based Evaluators through Multi-Agent Debate. *arXiv preprint arXiv:2308.07201* (2023). arXiv:2308.07201
- [12] Cheng et al. 2023. CoMPoS: Characterizing and Evaluating Caricature in LLM Simulations. In *EMNLP*. <https://doi.org/10.18653/v1/2023.emnlp-main.669>
- [13] Clarke and Dietz. 2024. LLM-based Relevance Assessment Still Can’t Replace Human Relevance Assessment. *arXiv preprint arXiv:2412.17156* (2024). arXiv:2412.17156
- [14] Dai et al. 2023. LLM-in-the-loop: Leveraging Large Language Model for Thematic Analysis. In *EMNLP-F*. <https://doi.org/10.18653/v1/2023.findings-emnlp.669>
- [15] Davison et al. 2024. The Ethics of Using Generative AI for Qualitative Data Analysis. *Information Systems J.* (2024). <https://doi.org/10.1111/isj.12504>
- [16] Duan et al. 2024. UICrit: Enhancing Automated Design Evaluation with a UI Critique Dataset. *arXiv preprint arXiv:2407.08850* (2024). arXiv:2407.08850
- [17] Dunivin. 2024. Scalable Qualitative Coding with LLMs: Chain-of-Thought Reasoning Matches Human Performance in Some Hermeneutic Tasks. *arXiv preprint arXiv:2401.15170* (2024). arXiv:2401.15170
- [18] Alexander R. Fabbri, Wojciech Kryściński, Bryan McCann, Caiming Xiong, Richard Socher, and Dragomir Radev. 2021. SummEval: Re-evaluating Summarization Evaluation. *Transactions of the Association for Computational Linguistics* 9 (2021), 391–409.

- [19] Gao et al. 2024. CollabCoder: A Lower-barrier, Rigorous Workflow for Inductive Collaborative Qualitative Analysis with Large Language Models. *arXiv preprint arXiv:2304.07366* (2024). arXiv:2304.07366
- [20] Gilardi et al. 2023. ChatGPT Outperforms Crowd Workers for Text-Annotation Tasks. *arXiv preprint arXiv:2303.15056* (2023). arXiv:2303.15056
- [21] Gu et al. 2024. A Survey on LLM-as-a-Judge. *arXiv preprint arXiv:2411.15594* (2024). arXiv:2411.15594
- [22] Guerino et al. 2025. Can GPT-4o Evaluate Usability Like Human Experts? A Comparative Study on Issue Identification in Heuristic Evaluation. *arXiv preprint arXiv:2506.16345* (2025). arXiv:2506.16345
- [23] Hashemi et al. 2024. LLM-Rubric: A Multidimensional, Calibrated Approach to Automated Evaluation of Natural Language Texts. *arXiv preprint arXiv:2501.00274* (2024). arXiv:2501.00274
- [24] Haxvig et al. 2025. I've Never Seen a Glass Ceiling Better Represented: Bias and Gendering in LLM-Generated Synthetic Personas from a Participatory Design Perspective. *Int. J. Hum.-Comput. Stud.* (2025). <https://doi.org/10.1016/j.ijhcs.2025.103651>
- [25] Huang et al. 2025. How Many Human Survey Respondents is a Large Language Model Worth? An Uncertainty Quantification Perspective. *arXiv preprint arXiv:2502.17773* (2025). arXiv:2502.17773
- [26] Jain et al. 2025. Multi-LLM Thematic Analysis with Dual Reliability Metrics: Combining Cohen's Kappa and Semantic Similarity for Qualitative Research Validation. *arXiv preprint arXiv:2512.20352* (2025). arXiv:2512.20352
- [27] James et al. 2026. Counting on Consensus: Selecting the Right Inter-annotator Agreement Metric for NLP Annotation and Evaluation. *arXiv preprint arXiv:2603.06865* (2026). arXiv:2603.06865
- [28] Kim et al. 2023. Prometheus: Inducing Fine-grained Evaluation Capability in Language Models. *arXiv preprint arXiv:2310.08491* (2023). arXiv:2310.08491
- [29] Kocmi and Federmann. 2023. GEMBA-MQM: Detecting Translation Quality Error Spans with GPT-4. *arXiv preprint arXiv:2310.13988* (2023). arXiv:2310.13988
- [30] Li et al. 2024. Comparing GPT-4 and Human Researchers in Health Care Data Analysis: Qualitative Description Study. *J. Med. Internet Res.* (2024). <https://doi.org/10.2196/56500>
- [31] Li et al. 2024. From Generation to Judgment: Opportunities and Challenges of LLM-as-a-judge. *arXiv preprint arXiv:2411.16594* (2024). arXiv:2411.16594
- [32] Li et al. 2024. LLMs-as-Judges: A Comprehensive Survey on LLM-based Evaluation Methods. *arXiv preprint arXiv:2412.05579* (2024). arXiv:2412.05579
- [33] Liu et al. 2023. G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment. *arXiv preprint arXiv:2303.16634* (2023). arXiv:2303.16634
- [34] Nora McDonald, Sarita Schoenebeck, and Andrea Forte. 2019. Reliability and Inter-rater Reliability in Qualitative Research: Norms and Guidelines for CSCW and HCI Practice. In *Proceedings of the ACM on Human-Computer Interaction (CSCW)*, Vol. 3. ACM, 1–23.
- [35] Meng et al. 2024. Exploring the Human-LLM Synergy in Advancing Theory-driven Qualitative Analysis (CHALET). *arXiv preprint arXiv:2405.05758* (2024). arXiv:2405.05758
- [36] Ngo et al. 2026. Qualitative Coding Analysis through Open-Source Large Language Models: A User Study and Design Recommendations. *arXiv preprint arXiv:2602.18352* (2026). arXiv:2602.18352
- [37] Nguyen-Trung. 2025. ChatGPT in Thematic Analysis: Can AI Become a Research Assistant in Qualitative Research? *Quality & Quantity* (2025). <https://doi.org/10.1007/s11135-025-02165-z>
- [38] Matthew J. Page, Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, et al. 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 372 (2021), n71. <https://doi.org/10.1136/bmj.n71>
- [39] Pangakis et al. 2023. Automated Annotation with Generative AI Requires Validation. *arXiv preprint arXiv:2306.00176* (2023). arXiv:2306.00176
- [40] De Paoli. 2023. Performing an Inductive Thematic Analysis of Semi-Structured Interviews with a Large Language Model. *Soc. Sci. Comput. Rev.* (2023). <https://doi.org/10.1177/08944393231220483>
- [41] Barbara Plank. 2022. The "Problem" of Human Label Variation: On Ground Truth in Data, Modeling and Evaluation. *Proceedings of EMNLP* (2022), 10671–10682.
- [42] Salminen et al. 2024. Deus Ex Machina and Personas from Large Language Models: Investigating the Composition of AI-Generated Persona Descriptions. In *CHI*. <https://doi.org/10.1145/3613904.3642036>
- [43] Salminen et al. 2025. The Use of Large Language Models in HCI: A Critical Analysis of Synthetic Users. *Augmented Humans* (2025). <https://doi.org/10.1145/3745900.3746108>
- [44] Schroeder et al. 2025. Large Language Models in Qualitative Research: Uses, Tensions, and Intentions. In *CHI*. <https://doi.org/10.1145/3706598.3713120>
- [45] Tornberg. 2023. ChatGPT-4 Outperforms Experts and Crowd Workers in Annotating Political Twitter Messages with Zero-Shot Learning. *arXiv preprint arXiv:2304.06588* (2023). arXiv:2304.06588
- [46] Verga et al. 2024. Replacing Judges with Juries: Evaluating LLM Generations with a Panel of Diverse Models. *arXiv preprint arXiv:2404.18796* (2024). arXiv:2404.18796
- [47] Wang et al. 2023. PandaLM: An Automatic Evaluation Benchmark for LLM Instruction Tuning Optimization. *arXiv preprint arXiv:2306.05087* (2023). arXiv:2306.05087
- [48] Wang et al. 2024. Large Language Models that Replace Human Participants can Harmfully Misportray and Flatten Identity Groups. *arXiv preprint arXiv:2402.01908* (2024). arXiv:2402.01908
- [49] Wei et al. 2024. Long-form Factuality in Large Language Models. *arXiv preprint arXiv:2403.18802* (2024). arXiv:2403.18802
- [50] Xiao et al. 2023. Supporting Qualitative Analysis with Large Language Models: Combining Codebook with GPT-3 for Deductive Coding. *arXiv preprint arXiv:2304.10548* (2023). arXiv:2304.10548

- [51] Xiao et al. 2025. Qualitative Coding with GPT-4: Where it Works Better. *J. Learning Analytics* (2025).
- [52] Zheng et al. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *arXiv preprint arXiv:2306.05685* (2023). [arXiv:2306.05685](#)
- [53] Zhong et al. 2025. Synthetic Heuristic Evaluation: A Comparison between AI- and Human-Powered Usability Evaluation. *arXiv preprint arXiv:2507.02306* (2025). [arXiv:2507.02306](#)
- [54] Zhu et al. 2023. JudgeLM: Fine-tuned Large Language Models are Scalable Judges. *arXiv preprint arXiv:2310.17631* (2023). [arXiv:2310.17631](#)

A REVIEW PROTOCOL AND RELIABILITY RUBRIC

Identification. Both reviews follow PRISMA 2020. The primary bibliographic source is the **OpenAlex** API; we cross-check with the **Semantic Scholar** Graph API and a structured web “**deep-research**” arm. Study 1a used a 59-query web protocol plus venue-scoped sweeps; Study 2 used a 25-query HCC/HCI/qualitative concept-block protocol (the same query strings were issued against every source so the arms are comparable). Concept blocks combine an LLM term (“LLM”, “large language model”, “GPT-4”, “ChatGPT”, “Gemini”), an evaluation/agreement term (“LLM-as-a-judge”, “automatic evaluation”, “annotation”, “inter-annotator agreement”, “human evaluation”), and, for Study 2, an HCC method term (“qualitative coding”, “thematic analysis”, “content analysis”, “crowdsourcing”, “participant simulation”, “persona”, “usability evaluation”, “design critique”). Records were deduplicated by DOI and normalized title.

Screening and eligibility. *Inclusion:* peer-reviewed or preprint studies that empirically compare LLM judgments to human judgments on the same items and report (or allow computing) an agreement statistic. *Exclusion:* non-empirical position papers, studies with no human baseline, studies that use LLMs only to generate (not judge) content, and off-topic retrievals. Title/abstract screening was followed by full-text eligibility assessment; per-stage counts appear in the PRISMA flow figures (Figures 2 and 6).

Reliability verdict (five-level rubric). Each study is coded against the *human-human agreement ceiling* for its task:

- **1 Validated:** LLM-human agreement is at or above the human-human ceiling (substitution is empirically justified).
- **2 Promising:** agreement approaches the ceiling under specific, reported conditions (model, prompt, aggregation); conditional substitution.
- **3 Mixed:** results are task- or setup-dependent and inconsistent across studies.
- **4 Caution:** agreement is consistently and materially below the ceiling.
- **5 Unreliable:** agreement is near chance or dominated by bias (position, verbosity, self-preference).

We report a reliability score = 6 – verdict and synthesize within metric families (e.g. Cohen’s κ , Krippendorff’s α , Spearman ρ , accuracy) rather than pooling incommensurable effect sizes. The full coded corpora (Appendices D and E) are released for re-coding.

B STUDY 1B: G-EVAL REPLICATION DETAILS

We replicate the summary-level G-Eval protocol [33] on SummEval [18] (100 source articles $\times$ 16 system summaries = 1600 summaries, each with expert means for four aspects). Each (summary, aspect) pair is scored 1 to 5 by the judge with the aspect criterion below; we then compute the **summary-level Spearman ρ** (per source document, correlate the judge’s 16 scores with the human means, averaged over documents, dropping documents with no score variance) and compare it to the reported G-Eval-4/GPT-4 correlations. Judges are the four most powerful open models in the available open-model catalogue (g1m-5.1-fp8 754B, deepseek-v4-pro, gptoss-120b, qwen3.6-35b-a3b-fp8), decoded at temperature 0. Aspect criteria (verbatim):

Coherence (1-5): the collective quality of all sentences; the summary should be well-structured and well-organized, building from sentence to sentence into a coherent body of information, not just a heap of related information.

Consistency (1-5): factual alignment between the summary and the source; a consistent summary contains only statements entailed by the source document; hallucinated or unsupported facts are penalized.

Fluency (1-5): the quality of individual sentences (grammar, capitalization, readability).

Relevance (1-5): selection of the most important content from the source; redundancy and unimportant content are penalized.

The judge receives the source article and the summary and returns only a JSON object {"score": <1-5>} (a larger token budget and a retry handle reasoning-model outputs). In total Study 1b contributes 25,600 judgments.

C STUDY 3: DATASETS, JUDGING PROMPTS, AND METRICS

Datasets. Table 5 summarizes the six datasets. Each is sampled to $n=500$ items with a clear majority label (ties dropped); we use the per-item annotator distributions to compute the human ceiling.

Table 5. Study 3 dataset cards. Ceiling = random-annotator-vs-majority accuracy. Contam. = max verbatim-recall tier across probed judges.

Dataset	Task	Source	Annot./item	Ceiling	Contam.
ChaosNLI-SNLI	NLI (3-way)	SNLI + ChaosNLI relabels	~100	0.75	MEDIUM
ChaosNLI-MNLI	NLI (3-way)	MultiNLI + ChaosNLI relabels	~100	0.64	MEDIUM
GoEmotions	sentiment (4-way)	Reddit (Google)	~3-5	0.78	MEDIUM
SBIC	offensiveness (binary)	Social Bias Frames	~3	0.91	MEDIUM
HateXplain	hate speech (3-way)	Twitter/Gab	3	0.83	MEDIUM
MultiPiCo	irony (binary)	Reddit/Twitter (2024)	3-8	0.82	LOW/MED

Judging prompts. Each model judges every item as a virtual annotator under instructions paralleling the human guidelines, returning a JSON label on the final line (parsed with a surface-form-to-code map). The exact user prompts:

NLI: Premise: {premise} / Hypothesis: {hypothesis} / Does the premise ENTAIL the hypothesis, CONTRADICT it, or is it NEUTRAL? -> {"label": "entailment"|"neutral"|"contradiction"}

Hate: Post: {text} / Classify: HATE SPEECH (attacks/dehumanizes a group based on identity), OFFENSIVE (abusive but not group-targeted), or NORMAL. -> {"label": "hate speech"|"offensive"|"normal"}

Emotion: Comment: {text} / Overall sentiment: POSITIVE, NEGATIVE, AMBIGUOUS (mixed/uncertain), or NEUTRAL. -> {"label": "positive"|"negative"|"ambiguous"|"neutral"}

Offensive: Post: {text} / Is this post offensive (including potentially offensive)? -> {"label": "offensive"|"not offensive"}

Irony: {post+reply} / Is the REPLY ironic/sarcastic (conveys a meaning different from or opposite to its literal content), or not ironic? -> {"label": "ironic"|"not ironic"}

Metrics. Let an item have A human annotations with majority label y^* . **Human ceiling** is the random-annotator-vs-majority accuracy (the expected agreement of one held-out human with the majority), averaged over items; we also report the pairwise human-human agreement. **Accuracy**, **Cohen’s κ** , and **macro-F1** are computed against y^* . The **alt-test** winning rate [7] is the fraction of annotators for whom replacing that human with the LLM does not reduce agreement with the remaining humans (a count-based leave-one-annotator-out test); a winning rate ≥ 0.5 is the

bar for claiming the LLM can “replace” a human. The **jury** is the per-item majority vote of the model panel [46]. **95% confidence intervals** are nonparametric bootstrap (1000 item resamples). **Power:** at $n=500$ the accuracy CI half-width is $\leq \pm 4.4\%$ and the design has $> 80\%$ power to detect a 0.10 gap from the ceiling. **Contamination probe:** the judge is given the dataset name and a prefix ($\approx$ one third) of an item’s text and asked to reproduce the remainder verbatim; we report the mean normalized sequence overlap and flag HIGH (> 0.6 mean or $> 20\%$ near-verbatim), MEDIUM (> 0.35 mean), or LOW. Only item-level labels and aggregates are released (no raw text); the runs recorded *zero* API errors and a 100% answer-parse rate across all 18000 judgments.

Inferential model. The mixed-effects analysis models each judgment’s correctness ($1[\text{judge} = y^*]$) with fixed effects for task subjectivity (1–ceiling, dataset level) and model scale ($\log_{10}$ parameters), reported as a Bayesian logistic GLMM with random intercepts for dataset and item and, for clean inference, as a logistic GLM with item-cluster-robust standard errors; the Benjamini-Hochberg procedure controls the false-discovery rate across the 36 judge $\times$ dataset ceiling-gap tests.

D STUDY 1 CORPUS

Table 6. Full extracted-studies corpus (217 included studies). Agreement is the headline coefficient/proportion where reported.

Key	Study	Domain	Task	Metric	Agree	Verdict
mirzakhmedova2024argument	Mirzakhmedova et al. (2024)	Argument Quality	pointwise	pct_agreement	-	Mixed
tong2024codejudge	Tong et al. (2024)	Code & SW Eng	classify	accuracy	-	Promising
watson2024aicodereview	Watson et al. (2024)	Code & SW Eng	classify	pct_agreement	-	Mixed
codespec2025	Jin et al. (2025)	Code & SW Eng	verify	accuracy	-	Caution
wang2025setesting	Wang et al. (2025)	Code & SW Eng	pairwise	pct_agreement	-	Mixed
zhuo2025codejudgeeff	Wang et al. (2025)	Code & SW Eng	pointwise	kappa	-	Mixed
biasloop2026	Zhao et al. (2026)	Code & SW Eng	pairwise	pct_agreement	-	Caution
overcorrect2026	Jin et al. (2026)	Code & SW Eng	classify	accuracy	-	Caution
hatespeechprobe2023	Roy et al. (2023)	Content Moderation & Toxicity	classify	f1	-	Mixed
li2023hotchatgpt	Li et al. (2023)	Content Moderation & Toxicity	classify	pct_agreement	-	Mixed
nguyen2024crisismisinfo	Nguyen & Rudra (2024)	Content Moderation & Toxicity	classify	f1	-	Mixed
hatespeechanno2025	authors (2025)	Content Moderation & Toxicity	classify	kappa	-	Caution
icwsm2025hatebias	Giorgi et al. (2025)	Content Moderation & Toxicity	classify	kappa	-	Caution
xie2023nextchapter	Xie et al. (2023)	Creative Writing	pointwise	pct_agreement	-	Mixed
chakrabarty2024artartifice	Chakrabarty et al. (2024)	Creative Writing	pointwise	pearson	0.10	Unreliable
ismayilzada2024creative	Ismayilzada et al. (2024)	Creative Writing	pointwise	pct_agreement	-	Caution
creativeevalmdpi2025	Kim et al. (2025)	Creative Writing	pointwise	spearman	-	Mixed
haase2025creativitypeaked	Haase et al. (2025)	Creative Writing	pointwise	pct_agreement	-	Caution
litbench2025	Fein et al. (2025)	Creative Writing	pairwise	pct_agreement	0.78	Mixed
lin2023llmeval	Lin & Chen (2023)	Dialogue	pointwise	spearman	-	Promising
mendonca2024dialogue	Mendonca et al. (2024)	Dialogue	pointwise	spearman	-	Mixed
hackl2023reliablerater	Hackl et al. (2023)	Education & Grading	pointwise	icc	-	Promising
khademi2023bardassessment	Khademi (2023)	Education & Grading	pointwise	pct_agreement	-	Mixed
kortemeyer2023physics	Kortemeyer (2023)	Education & Grading	pointwise	pct_agreement	-	Mixed
mizumoto2023aes	Mizumoto & Eguchi (2023)	Education & Grading	pointwise	qwk	0.74	Promising
yancey2023cefr	Yancey et al. (2023)	Education & Grading	pointwise	qwk	-	Mixed
aescomparative2024	Lee et al. (2024)	Education & Grading	pointwise	qwk	0.57	Mixed
floden2024gradingexams	Flodén (2024)	Education & Grading	pointwise	pct_agreement	-	Mixed
grevisse2024asagmed	Grévisse (2024)	Education & Grading	classify	pct_agreement	-	Promising
hou2024classroom	Hou et al. (2024)	Education & Grading	pointwise	icc	-	Mixed
k12shortanswer2024	Henkel et al. (2024)	Education & Grading	classify	kappa	0.70	Promising

Key	Study	Domain	Task	Metric	Agree	Verdict
koraishi2024langassess	Koraishi (2024)	Education & Grading	pointwise	pct_agreement	-	Mixed
quah2024dentalaes	Quah et al. (2024)	Education & Grading	pointwise	icc	-	Mixed
wang2024beyondagreement	Wang et al. (2024)	Education & Grading	pointwise	qwk	-	Mixed
writingmultidim2024	Steiss et al. (2024)	Education & Grading	pointwise	qwk	-	Promising
asagfair2025	Henkel et al. (2025)	Education & Grading	pointwise	kappa	-	Mixed
sefl2025feedback	authors (2025)	Education & Grading	pointwise	f1	-	Promising
studentsjudge2025	Banihashem et al. (2025)	Education & Grading	pointwise	pct_agreement	-	Mixed
k12sci2026	He et al. (2026)	Education & Grading	pointwise	pct_agreement	-	Promising
claimmatch2023	Choi et al. (2023)	Fact-Checking	classify	pct_agreement	-	Promising
ni2024afacta	Ni et al. (2024)	Fact-Checking	classify	pct_agreement	-	Promising
politicaltruths2024	Chatrath et al. (2024)	Fact-Checking	classify	pct_agreement	-	Mixed
factsopinions2025	Saju et al. (2025)	Fact-Checking	classify	accuracy	-	Caution
kaikaus2023financecorpora	Kaikaus et al. (2023)	Finance	classify	accuracy	-	Mixed
bizfinbench2025	Lu et al. (2025)	Finance	pointwise	pct_agreement	-	Mixed
trident2025	Hui et al. (2025)	Finance	classify	pct_agreement	-	Mixed
chan2023chateval	Chan et al. (2023)	General NLG	pairwise	pct_agreement	-	Mixed
chiang2023alt	Chiang & Lee (2023)	General NLG	pointwise	pct_agreement	-	Mixed
dubois2023alpacaform	Dubois et al. (2023)	General NLG	pairwise	winrate_corr	0.98	Promising
hsu2023figurecaption	Hsu et al. (2023)	General NLG	pointwise	spearman	-	Promising
kim2023prometheus	Kim et al. (2023)	General NLG	pointwise	pearson	0.90	Promising
lai2023styletransfer	Lai et al. (2023)	General NLG	pointwise	spearman	-	Mixed
li2023autoj	Li et al. (2023)	General NLG	pairwise	pct_agreement	-	Mixed
wang2023chatgptnl	Wang et al. (2023)	General NLG	pointwise	spearman	-	Mixed
wang2023pandal	Wang et al. (2023)	General NLG	pairwise	accuracy	-	Promising
xu2023instructscore	Xu et al. (2023)	General NLG	pointwise	pearson	-	Promising
zhang2023widerdeeper	Zhang et al. (2023)	General NLG	pairwise	pct_agreement	-	Mixed
zheng2023mtbench	Zheng et al. (2023)	General NLG	pairwise	pct_agreement	0.85	Validated
chiang2024arena	Chiang et al. (2024)	General NLG	pairwise	pct_agreement	-	Validated
doostmohammadi2024autoeval	Doostmohammadi et al. (2024)	General NLG	pointwise	pct_agreement	-	Mixed
dubois2024lc	Dubois et al. (2024)	General NLG	pairwise	winrate_corr	0.98	Promising
hashemi2024llmrubric	Hashemi et al. (2024)	General NLG	pointwise	pearson	-	Promising
huang2024nlexpl	Huang et al. (2024)	General NLG	pointwise	pct_agreement	-	Mixed
kim2024prometheus2	Kim et al. (2024)	General NLG	pointwise	pearson	-	Promising
li2024arenahard	Li et al. (2024)	General NLG	pairwise	pct_agreement	0.89	Promising
lin2024wildbench	Lin et al. (2024)	General NLG	pairwise	winrate_corr	0.98	Promising
verga2024poll	Verga et al. (2024)	General NLG	pairwise	pct_agreement	-	Promising
faggioli2023perspectives	Faggioli et al. (2023)	IR & Relevance	classify	kappa	-	Promising
alaofi2024fooledrelevant	Alaofi et al. (2024)	IR & Relevance	classify	pct_agreement	-	Caution
clarke2024cantreplace	Clarke & Dietz (2024)	IR & Relevance	classify	kappa	-	Caution
thomas2024relevance	Thomas et al. (2024)	IR & Relevance	classify	kappa	-	Validated
upadhyay2024largescale	Upadhyay et al. (2024)	IR & Relevance	classify	kappa	-	Promising
zhuang2024beyondyesno	Zhuang et al. (2024)	IR & Relevance	ranking	kendall	-	Promising
arabzadeh2025benchrelevance	Arabzadeh & Clarke (2025)	IR & Relevance	classify	kappa	-	Mixed
arabzadeh2025promptsens	Arabzadeh & Clarke (2025)	IR & Relevance	classify	kappa	-	Mixed
frobe2025assessorsagree	Frobe et al. (2025)	IR & Relevance	classify	kappa	-	Caution
li2023coannotating	Li et al. (2023)	Information Extraction (NER/NLI)	classify	pct_agreement	-	Promising
humanllmcollab2024	Wang et al. (2024)	Information Extraction (NER/NLI)	classify	pct_agreement	-	Promising
kholodna2024activelearning	Kholodna et al. (2024)	Information Extraction (NER/NLI)	classify	f1	-	Promising
kim2024meganno	Kim et al. (2024)	Information Extraction (NER/NLI)	classify	f1	-	Promising
nasution2024chatgptlabel	Nasution & Onan (2024)	Information Extraction (NER/NLI)	classify	accuracy	-	Mixed
ner2024augment	Naraki et al. (2024)	Information Extraction (NER/NLI)	classify	f1	-	Promising
nli2024labeldist	Chen et al. (2024)	Information Extraction (NER/NLI)	classify	pearson	-	Promising
ronningstad2024gptannotator	Ronningstad et al. (2024)	Information Extraction (NER/NLI)	classify	f1	-	Mixed
qiu2025labelensemble	Qiu et al. (2025)	Information Extraction (NER/NLI)	classify	accuracy	-	Promising

	Key	Study	Domain	Task	Metric	Agree	Verdict
1197							
1198	zhou2023ifeval	Zhou et al. (2023)	Instruction Following	verify	accuracy	-	Validated
1199	zeng2024llmbar	Zeng et al. (2024)	Instruction Following	pairwise	accuracy	-	Mixed
1200	ifcritic2025	Wen et al. (2025)	Instruction Following	classify	pct_agreement	-	Promising
1201	legaldocrec2025	Pradhan et al. (2025)	Legal	ranking	spearman	0.73	Promising
1202	lemaj2025	Enguehard et al. (2025)	Legal	pointwise	pct_agreement	-	Mixed
1203	kocmi2023gemba	Kocmi & Federmann (2023)	Machine Translation	pointwise	kendall	-	Promising
1204	kocmi2023genbamqm	Kocmi & Federmann (2023)	Machine Translation	verify	pct_agreement	0.96	Validated
1205	lu2023erroranalysis	Lu et al. (2023)	Machine Translation	verify	kendall	-	Promising
1206	sun2023radimpressions	Sun et al. (2023)	Medicine & Clinical	pointwise	pct_agreement	-	Caution
1207	tang2023medevidence	Tang et al. (2023)	Medicine & Clinical	pointwise	pct_agreement	-	Caution
1208	brain2024radgpt	Mitsuyama et al. (2024)	Medicine & Clinical	classify	accuracy	0.94	Mixed
1209	brake2024clinicalnote	Brake & Schaaf (2024)	Medicine & Clinical	pointwise	pct_agreement	-	Mixed
1209	wang2024clinicalevent	Wang et al. (2024)	Medicine & Clinical	classify	pct_agreement	-	Promising
1210	chung2025verifact	Chung et al. (2025)	Medicine & Clinical	verify	pct_agreement	-	Mixed
1211	clinstruct2025	Brake et al. (2025)	Medicine & Clinical	pointwise	icc	0.82	Mixed
1212	croxford2025autoeval	Croxford et al. (2025)	Medicine & Clinical	pointwise	pct_agreement	-	Mixed
1213	croxford2025clinsummjudge	Croxford et al. (2025)	Medicine & Clinical	pointwise	pct_agreement	-	Mixed
1214	healthbench2025	Arora et al. (OpenAI) (2025)	Medicine & Clinical	pointwise	pct_agreement	-	Promising
1215	kim2025emulate	Kim & Kim (2025)	Medicine & Clinical	pointwise	pct_agreement	-	Mixed
1216	llmevalmed2025	Zhang et al. (2025)	Medicine & Clinical	pointwise	pct_agreement	-	Mixed
1217	szymanski2025limits	Szymanski et al. (2025)	Medicine & Clinical	pairwise	pct_agreement	0.66	Caution
1218	clindialog2026	Sun et al. (2026)	Medicine & Clinical	pointwise	pct_agreement	-	Mixed
1219	frenchmedqa2026	Belmadani et al. (2026)	Medicine & Clinical	pointwise	kappa	-	Mixed
1219	medjudge2026scoping	Li et al. (2026)	Medicine & Clinical	review	nan	-	Caution
1220	counselbench2025	Li et al. (2025)	Mental Health	pointwise	pct_agreement	-	Caution
1221	hua2025mhscoping	Hua et al. (2025)	Mental Health	review	nan	-	Caution
1222	mhtrust2025	Badawi et al. (2025)	Mental Health	pointwise	icc	-	Mixed
1223	mhsupport2026	Badawi et al. (2026)	Mental Health	pointwise	icc	-	Mixed
1224	liu2023trustworthy	Liu et al. (2023)	Methodology & Bias	survey	nan	-	Mixed
1225	saito2023verbositybias	Saito et al. (2023)	Methodology & Bias	pairwise	pct_agreement	-	Caution
1226	wang2023shepherd	Wang et al. (2023)	Methodology & Bias	pointwise	pct_agreement	-	Promising
1227	zhu2023judgelm	Zhu et al. (2023)	Methodology & Bias	pairwise	pct_agreement	-	Promising
1228	bavaresco2024judgebench	Bavaresco et al. (2024)	Methodology & Bias	classify	kappa	0.28	Mixed
1229	chen2024humansorllms	Chen et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Caution
1229	desmond2024evalullm	Desmond et al. (2024)	Methodology & Bias	pointwise	pct_agreement	-	Mixed
1230	fan2024sedareval	Fan et al. (2024)	Methodology & Bias	pointwise	pct_agreement	-	Promising
1231	feuer2024style	Feuer et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Caution
1232	gu2024survey	Gu et al. (2024)	Methodology & Bias	survey	nan	-	Mixed
1233	jung2024trustescalate	Jung et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Promising
1234	justice2024prejudice	Ye et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Caution
1235	ke2024critiquellm	Ke et al. (2024)	Methodology & Bias	pointwise	spearman	-	Mixed
1236	koo2024cobbler	Koo et al. (2024)	Methodology & Bias	pairwise	winrate_corr	0.44	Caution
1237	li2024compsurvey	Li et al. (2024)	Methodology & Bias	survey	nan	-	Mixed
1238	li2024gen2judge	Li et al. (2024)	Methodology & Bias	survey	nan	-	Mixed
1238	li2024splitmerge	Li et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Mixed
1239	liu2024pairwisepref	Liu et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Promising
1240	pan2024hcdjudge	Pan et al. (2024)	Methodology & Bias	pointwise	nan	-	Mixed
1241	panickssery2024selfpref	Panickssery et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Caution
1242	raina2024robust	Raina et al. (2024)	Methodology & Bias	pointwise	pct_agreement	-	Unreliable
1243	shankar2024validators	Shankar et al. (2024)	Methodology & Bias	pointwise	pct_agreement	-	Mixed
1244	stureborg2024inconsistent	Stureborg et al. (2024)	Methodology & Bias	pointwise	pct_agreement	-	Caution
1245	thakur2024judgingjudges	Thakur et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Mixed
1246	wataoka2024selfpref	Wataoka et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Caution
1247	zhu2024nationalitybias	Zhu et al. (2024)	Methodology & Bias	pointwise	pct_agreement	-	Caution
1248							

Key	Study	Domain	Task	Metric	Agree	Verdict
calderon2025alttest	Calderon et al. (2025)	Methodology & Bias	classify	pct_agreement	-	Promising
gao2025nlgsurvey	Gao et al. (2025)	Methodology & Bias	survey	nan	-	Mixed
gao2025reranking	Gao et al. (2025)	Methodology & Bias	ranking	kendall	-	Mixed
hu2025trainjudge	Hu et al. (2025)	Methodology & Bias	pointwise	pct_agreement	-	Promising
huang2025empiricaljudge	Huang et al. (2025)	Methodology & Bias	pointwise	pct_agreement	-	Mixed
judgesverdict2025	Han et al. (2025)	Methodology & Bias	pointwise	kappa	-	Promising
li2025prefleakage	Li et al. (2025)	Methodology & Bias	pairwise	pct_agreement	-	Caution
murugadoss2025evaltheeval	Murugadoss et al. (2025)	Methodology & Bias	pointwise	pct_agreement	-	Mixed
voutsas2025biaseddesign	Voutsas et al. (2025)	Methodology & Bias	classify	pct_agreement	-	Caution
yamauchi2025designchoices	Yamauchi et al. (2025)	Methodology & Bias	pointwise	pct_agreement	-	Mixed
imrit2026iaa	James et al. (2026)	Methodology & Bias	review	krippendorff_alpha	-	Mixed
temperature2026	Li et al. (2026)	Methodology & Bias	pointwise	pct_agreement	-	Mixed
khondaker2023gptaraeval	Khondaker et al. (2023)	Multilingual	classify	f1	-	Caution
mmeval2024	Son et al. (2024)	Multilingual	pairwise	accuracy	-	Mixed
multiljudge2025	Fu et al. (2025)	Multilingual	pointwise	pct_agreement	-	Caution
reliablemulti2026	Zubiaga et al. (2026)	Multilingual	pointwise	pct_agreement	-	Caution
chen2024mllmjudge	Chen et al. (2024)	Multimodal (Vision-Language)	pairwise	pct_agreement	0.79	Mixed
duan2024uimockup	Duan et al. (2024)	Multimodal (Vision-Language)	pointwise	pct_agreement	-	Mixed
lee2024promvision	Lee et al. (2024)	Multimodal (Vision-Language)	pointwise	pearson	-	Mixed
lu2024wildvision	Lu et al. (2024)	Multimodal (Vision-Language)	pairwise	pct_agreement	-	Promising
yasunaga2025mmrewardbench	Yasunaga et al. (2025)	Multimodal (Vision-Language)	pairwise	accuracy	0.72	Caution
chatgptreviewers2024	Ho et al. (2024)	Peer Review & Science	pointwise	pct_agreement	0.07	Caution
liang2024feedback	Liang et al. (2024)	Peer Review & Science	pointwise	pct_agreement	-	Mixed
murad2024quality	Tarakji et al. (2024)	Peer Review & Science	classify	pct_agreement	0.61	Mixed
prismascreen2024	Guo et al. (2024)	Peer Review & Science	classify	kappa	0.91	Validated
rose2025rob	Rose et al. (2025)	Peer Review & Science	classify	kappa	-	Mixed
shopovski2025peerreview	Shopovski et al. (2025)	Peer Review & Science	pointwise	pct_agreement	-	Mixed
guo2023showclose	Guo et al. (2023)	QA & Factuality	pointwise	pct_agreement	-	Mixed
min2023factscore	Min et al. (2023)	QA & Factuality	verify	pct_agreement	-	Promising
li2024pedants	Li et al. (2024)	QA & Factuality	verify	pct_agreement	-	Promising
madaan2024refverdict	Reference-Guided Verdict authors (2024)	QA & Factuality	pointwise	pct_agreement	-	Promising
wei2024safe	Wei et al. (2024)	QA & Factuality	verify	pct_agreement	0.72	Promising
leas2023contentanalysis	Leas et al. (2023)	Qualitative Coding	classify	pct_agreement	-	Mixed
meng2024chalet	Meng et al. (2024)	Qualitative Coding	classify	pct_agreement	-	Mixed
qualigpt2024	Zhang et al. (2024)	Qualitative Coding	classify	pct_agreement	-	Mixed
mehta2025qualgenai	Mehta et al. (2025)	Qualitative Coding	classify	pct_agreement	-	Mixed
qualcoding2025jla	Xiao et al. (2025)	Qualitative Coding	classify	fleiss_kappa	0.46	Mixed
thematic2025charity	Wen et al. (2025)	Qualitative Coding	classify	pct_agreement	-	Mixed
es2023ragas	Es et al. (2023)	RAG	pointwise	pct_agreement	-	Mixed
saadfalcon2023ares	Saad-Falcon et al. (2023)	RAG	classify	accuracy	-	Promising
jin2024ragreward	Jin et al. (2024)	RAG	pairwise	accuracy	-	Mixed
raju2024calc2adj	Raju et al. (2024)	Reasoning & Math	verify	accuracy	-	Mixed
tan2024judgebench	Tan et al. (2024)	Reasoning & Math	pairwise	accuracy	0.55	Caution
mathrobust2026	Yosef et al. (2026)	Reasoning & Math	verify	accuracy	-	Mixed
lee2023rlaif	Lee et al. (2023)	Reward Modeling	pairwise	pct_agreement	-	Mixed
lambert2024rewardbench	Lambert et al. (2024)	Reward Modeling	pairwise	accuracy	-	Mixed
wu2024metarewarding	Wu et al. (2024)	Reward Modeling	pairwise	pct_agreement	-	Mixed
yuan2024selfrewarding	Yuan et al. (2024)	Reward Modeling	pairwise	pct_agreement	-	Mixed
ifreward2026	Wen et al. (2026)	Reward Modeling	ranking	kendall	0.61	Caution
belal2023sentiment	Belal et al. (2023)	Sentiment & Emotion	classify	accuracy	-	Promising
koptyra2023clarinemo	Koptyra et al. (2023)	Sentiment & Emotion	classify	pct_agreement	-	Mixed
zhang2023sentiment	Zhang et al. (2023)	Sentiment & Emotion	classify	accuracy	-	Mixed
niu2024text2emotion	Niu et al. (2024)	Sentiment & Emotion	classify	pct_agreement	-	Mixed
aldeen2023chatgptvshuman	Aldeen et al. (2023)	Social-Science Annotation	classify	accuracy	-	Mixed

Key	Study	Domain	Task	Metric	Agree	Verdict
cegin2023paraphrases	Cegin et al. (2023)	Social-Science Annotation	classify	pct_agreement	-	Mixed
gilardi2023chatgpt	Gilardi et al. (2023)	Social-Science Annotation	classify	accuracy	-	Validated
ollion2023mindhype	Ollion et al. (2023)	Social-Science Annotation	classify	pct_agreement	-	Caution
ostyakova2023crowdexperts	Ostyakova et al. (2023)	Social-Science Annotation	classify	pct_agreement	-	Mixed
pangakis2023validation	Pangakis et al. (2023)	Social-Science Annotation	classify	f1	0.71	Mixed
reiss2023testing	Reiss (2023)	Social-Science Annotation	classify	pct_agreement	-	Caution
wu2023llmworkers	Wu et al. (2023)	Social-Science Annotation	classify	pct_agreement	-	Mixed
zhu2023reproduce	Zhu et al. (2023)	Social-Science Annotation	classify	pct_agreement	-	Mixed
li2024stance	Li & Conrad (2024)	Social-Science Annotation	classify	pct_agreement	-	Mixed
pangakis2024keephumans	Pangakis & Wolken (2024)	Social-Science Annotation	classify	f1	-	Promising
ziems2024css	Ziems et al. (2024)	Social-Science Annotation	classify	accuracy	-	Mixed
ahmad2025moralmachine	Ahmad & Takemoto (2025)	Social-Science Annotation	classify	pct_agreement	-	Mixed
horych2025promisespitfalls	Horych et al. (2025)	Social-Science Annotation	classify	f1	-	Mixed
tornberg2025political	Törnberg (2025)	Social-Science Annotation	classify	accuracy	-	Validated
gao2023humanlike	Gao et al. (2023)	Summarization	pointwise	spearman	-	Mixed
liu2023geval	Liu et al. (2023)	Summarization	pointwise	spearman	0.51	Promising
luo2023factinconsistency	Luo et al. (2023)	Summarization	verify	pct_agreement	-	Mixed
shen2023notyet	Shen et al. (2023)	Summarization	pointwise	spearman	-	Caution
lee2024unisumeval	Lee et al. (2024)	Summarization	pointwise	pct_agreement	-	Promising
song2024finesure	Song et al. (2024)	Summarization	verify	pct_agreement	-	Promising
tang2024tofueval	Tang et al. (2024)	Summarization	verify	pct_agreement	-	Caution
zhang2024newsummm	Zhang et al. (2024)	Summarization	pointwise	pct_agreement	-	Promising
wu2023hpsv2	Wu et al. (2023)	Text-to-Image	pointwise	accuracy	-	Promising
lin2024vqascore	Lin et al. (2024)	Text-to-Image	pointwise	pearson	-	Promising

E STUDY 2 CORPUS

Table 7. Study 2 corpus (55 studies): HCC/HCI/qualitative methods using LLMs vs human input.

Study	Method	Role	Community	Agree	Verdict
Chew et al. (2023)	Content analysis	augment	CSS	-	Promising
Leas et al. (2023)	Content analysis	augment	Health	-	Mixed
Long et al. (2024)	Content analysis	augment	Education	-	Mixed
Dunivin (2025)	Content analysis	augment	CSS	-	Promising
Anderson et al. (2024)	Creativity/ideation	augment	HCI	-	Caution
Gilardi et al. (2023)	Crowdsourced annotation	replace	CSS	-	Validated
Tornberg (2023)	Crowdsourced annotation	replace	CSS	-	Validated
Kuzman et al. (2023)	Crowdsourced annotation	replace	NLP	-	Mixed
Li et al. (2023)	Crowdsourced annotation	augment	NLP	-	Promising
Ostyakova et al. (2023)	Crowdsourced annotation	replace	NLP	-	Mixed
Tabone & de Winter (2023)	HCI research methods	augment	HCI	-	Mixed
Aubin Le Quere et al. (2024)	HCI research methods	augment	HCI	-	Mixed
Salminen et al. (2024)	Persona generation	augment	HCI	-	Mixed
Jung et al. (2025)	Persona generation	augment	HCI	-	Mixed
Haxvig et al. (2025)	Persona generation	replace	HCI	-	Caution
Zhang et al. (2023)	Qual analysis (general)	augment	Qual-methods	-	Mixed
Wachinger et al. (2024)	Qual analysis (general)	augment	Health	-	Mixed
Li et al. (2024)	Qual analysis (general)	augment	Health	0.40	Mixed
Mehta et al. (2025)	Qual analysis (general)	augment	Qual-methods	-	Mixed
Davison et al. (2024)	Qual analysis (meta)	augment	CSS	-	Caution
Schroeder et al. (2025)	Qual analysis (meta)	augment	HCI	-	Mixed
Zhang et al. (2024)	Qual analysis (tool)	augment	HCI	-	Mixed

Study	Method	Role	Community	Agree	Verdict
Meng et al. (2024)	Qual analysis (tool)	augment	HCI	-	Mixed
Xiao et al. (2023)	Qual coding (deductive)	augment	HCI	-	Promising
Savelka et al. (2023)	Qual coding (deductive)	augment	NLP	-	Mixed
Zhang et al. (2025)	Qual coding (deductive)	augment	Qual-methods	0.46	Mixed
authors (2025)	Qual coding (deductive)	augment	Qual-methods	0.46	Mixed
Gao et al. (2024)	Qual coding (inductive)	augment	HCI	-	Promising
Dunivin (2024)	Qual coding (inductive)	replace	Qual-methods	-	Promising
Ngo et al. (2026)	Qual coding (inductive)	augment	HCI	-	Mixed
Wu et al. (2023)	Social-computing annotation	replace	CSCW	-	Mixed
Zhu et al. (2023)	Social-computing annotation	replace	CSCW	-	Mixed
Rajashekar et al. (2024)	Social-computing annotation	augment	HCI	-	Mixed
Argyle et al. (2023)	Synthetic participants	simulate	CSS	-	Mixed
Bisbee et al. (2023)	Synthetic participants	replace	CSS	-	Caution
Cheng et al. (2023)	Synthetic participants	simulate	NLP	-	Caution
Wang et al. (2024)	Synthetic participants	replace	HCI	-	Caution
Bail (2024)	Synthetic participants	simulate	CSS	-	Mixed
Abbasiantaeb et al. (2024)	Synthetic participants	simulate	IR	-	Mixed
Huang et al. (2025)	Synthetic participants	simulate	CSS	-	Mixed
Salminen et al. (2025)	Synthetic participants	replace	HCI	-	Caution
De Paoli (2023)	Thematic analysis	augment	CSS	-	Mixed
Dai et al. (2023)	Thematic analysis	augment	NLP	-	Promising
Prescott et al. (2024)	Thematic analysis	augment	Health	-	Mixed
Nguyen-Trung (2025)	Thematic analysis	augment	Qual-methods	-	Mixed
Jain et al. (2025)	Thematic analysis	replace	Qual-methods	0.91	Promising
Castellanos et al. (2025)	Thematic analysis	augment	Health	0.80	Promising
Sharma et al. (2025)	Thematic analysis	augment	HCI	-	Promising
Wen et al. (2025)	Thematic analysis	augment	Qual-methods	-	Mixed
Kocaballi (2023)	UX/design critique	augment	HCI	-	Mixed
Duan et al. (2024)	UX/design critique	augment	HCI	-	Mixed
Duan et al. (2024)	UX/design critique	augment	HCI	-	Mixed
Duan et al. (2024)	UX/design critique	augment	HCI	-	Mixed
Guerino et al. (2025)	Usability/heuristic eval	replace	HCI	0.21	Caution
Zhong et al. (2025)	Usability/heuristic eval	replace	HCI	0.75	Mixed